Composite Structures 391 (2026) 120506

Contents lists available at ScienceDirect

Composite Structures

journal homepage: www.elsevier.com/locate/compstruct

# Experimental evaluation of 3D-DIC mode shapes for detecting small-scale delaminations in composites laminates

Matheus Teixeira de Moura Santos, Guilherme Ferreira Gomes *

*Mechanical Engineering Institute, Federal University of Itajubá (UNIFEI), Brazil*

ARTICLE INFO

*Keywords:*
Structural health monitoring
3D-DIC
Delamination
Composite laminates
Vibration-Based Damage Detection (VBDD)
Wavelet Transform
Mode shapes

ABSTRACT

Structural Health Monitoring (SHM) of laminated composite materials faces significant challenges in detecting internal defects, particularly delaminations, which are often imperceptible to visual inspection. While traditional vibration-based methods provide useful diagnostic data, they are inherently limited by their discrete, point-based measurement approach. To address this, this work proposes a global, non-contact damage detection framework based on High-Speed Three-Dimensional Digital Image Correlation (3D-DIC). The methodology involves capturing full-field Operational Deflection Shapes (ODS) under broadband excitation (1–800 Hz) and systematically evaluating the efficacy of conventional metrics against an advanced Wavelet Transform Mode Shape Curvature (WT-MSC) method using the Mexican Hat wavelet. Quantitative analysis of standard curvature methods revealed severe limitations due to experimental noise, yielding a mean Signal-to-Noise Ratio (SNR) of –0.27 and a mean ROC-AUC of 0.41. Conversely, the proposed WT-MSC approach effectively denoised the signal and isolated damage-induced singularities, successfully pinpointing internal delaminations as small as 6.35 mm without requiring a pristine baseline. Furthermore, results demonstrate that higher-order vibration modes exhibit significantly higher sensitivity to small-scale defects. These quantitative findings validate the integration of 3D-DIC with wavelet-based processing as a robust and highly sensitive strategy for the Non-Destructive Testing (NDT) of composite aerostructures.

## 1. Introduction

Composite materials have been widely adopted in the market, particularly within the engineering domain, due to their satisfactory mechanical properties and high strength-to-weight ratio. These characteristics have made them a subject of intense study across various disciplines. However, with the constant increase in the use of composite laminates in the aeronautical industry, a critical weakness often referred to as the "Achilles' heel" of composite materials—has been identified: delamination [1]. This phenomenon involves the loss of adhesion between the material's fibers, rendering load transfer impracticable.

To illustrate the ubiquity of this failure mode across different industries and scales, Fig. 1 presents real-world examples of skin-debonding in an aircraft tail wing and damage on a wind turbine blade, alongside a microscopic cross-sectional view detailing the physical separation between layers.

In the short term, delamination can reduce the load transfer capacity between laminate layers, favoring damage propagation and leading to premature failure. In the long term, this condition can expose load-bearing layers to environmental agents, such as humidity or contaminants [2]. Consequently, the early and precise detection of such damage is fundamental to ensuring structural safety.

To identify potential hidden damages, a traditional method widely employed is Visual Inspection (VI). This technique is constantly applied to monitor aeronautical structures because it is cost-effective, rapid, and easily scalable [3]. Nevertheless, VI presents significant disadvantages, including subjectivity and a lack of reliability. Furthermore, in cases where damage is not externally visible, the method becomes ineffective. To address these limitations, the field of Structural Health Monitoring (SHM) has developed techniques to determine the state of a structure with greater precision. These techniques often involve the application of sensors, such as strain gauges (capable of measuring deformation at a specific point), accelerometers, or ultrasound sensors.

However, the application of contact sensors inevitably results in increased weight a factor that can alter the expected behavior of lightweight structures. Moreover, for very large objects of study, such as the aforementioned aircraft, a vast number of sensors is required for a single measurement campaign, making this method expensive and complex to implement. To mitigate this, non-contact techniques, have been utilized, such as the Laser Doppler Vibrometer (LDV) [4].

A clear disadvantage of both point-sensor methods and traditional point-based LDV regarding VI is that measurement occurs at a single

* Corresponding author.
*E-mail address:* guilhermefergom@unifei.edu.br (G.F. Gomes).

https://doi.org/10.1016/j.compstruct.2026.120506
Received 2 December 2025; Received in revised form 19 April 2026; Accepted 25 May 2026
Available online 11 June 2026
0263-8223/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

**Acronyms**

| | |
|---|---|
| 3D-DIC | Three-Dimensional Digital Image Correlation |
| CWT | Continuous Wavelet Transform |
| DIC | Digital Image Correlation |
| FFT | Fast Fourier Transform |
| FRF | Frequency Response Function |
| LDV | Laser Doppler Vibrometer |
| MSC | Mode Shape Curvature |
| NDT | Non-Destructive Testing |
| ODS | Operational Deflection Shape |
| ROI | Region of Interest |
| SHM | Structural Health Monitoring |
| VBDD | Vibration-Based Damage Detection |
| VI | Visual Inspection |
| WT | Wavelet Transform |
| WT-MSC | Wavelet Transform - Mode Shape Curvature |

**Symbols**

| | |
|---|---|
| $E$ | Young's Modulus |
| $I$ | Area Moment of Inertia |
| $M$ | Bending Moment |
| $s$ | Wavelet Scale Parameter |
| $h$ | Grid Spacing/Spatial Step |
| $\mathbf{b}$ | Translation Parameter Vector |
| $\mathbf{x}$ | Position Vector |
| $\boldsymbol{\Phi}$ | Modal Displacement/Mode Shape |
| $\boldsymbol{\Phi}''$ | Modal Curvature |
| $\sigma$ | Standard Deviation |
| $\psi$ | Wavelet Function (Mother Wavelet) |
| $\varphi$ | Scaling/Gaussian Function |
| $\theta$ | Rotation Angle |
| $\nabla^2$ | Laplacian Operator |

point on the structure. This necessitates multiple tests to achieve a result that covers a relevant portion of larger structures [5,6]. Consequently, a technique known as Vibration-Based Damage Detection (VBDD) was developed. This technique stands out as a fast, non-contact, and global inspection alternative.

VBDD is a method proposed to identify internal damage through differences in vibration response caused by structural defects. It relies on variations in parameters such as natural frequency, mode shapes, Modal Strain Energy (MSE), or the Frequency Response Function (FRF) [7]. This approach is particularly useful given that vibration data is relatively easy and inexpensive to acquire. However, while it is possible to identify the presence of damage through mode shape differences, the localization of said damage often requires the application of more complex methods based on multiple derivatives and specific transforms.

Many of the previously mentioned methods have applications in VBDD, such as accelerometers and LDVs. However, to satisfy the need for a global, non-contact analysis, a viable alternative has emerged: High-speed Three-Dimensional Digital Image Correlation (3D-DIC). This technique enables rapid, global, and non-contact analysis [8,9].

DIC relies on a sequence of images that capture the displacement of a stochastic pattern (speckles) painted on the surface of the structure being analyzed. This analysis is performed by dividing the image into sub-areas called subsets, where the displacement is determined by correlating these subsets before and after deformation.

The VBDD process can be approached primarily in two ways: physics-based or data-based. The physics-based approach uses numerical models to determine the body's behavior (e.g., determining the presence of damage through a discrepancy when compared to finite element models). Conversely, the data-based approach utilizes sensor data compared against a baseline structure, where any behavioral alteration may indicate an anomaly. Since a pristine baseline structure is not always available (as is often the case with large and complex structures in the aeronautical industry) a method that does not require comparison with a previous model, known as a baseline-free method, is highly desirable.

Given the necessity for a global, non-contact method, this work proposes the utilization of 3D-DIC combined with advanced post-processing methods for the detection and identification of delaminations in laminated composite structures. This identification strategy offers advantages for large-scale analyses, allows for a rapid evaluation of the total structure without the need for extensive instrumentation, and represents an innovative proposal for the non-destructive testing (NDT) sector.

The primary objective of this research is to investigate and evaluate the application of Digital Image Correlation (DIC) for detecting multiple delamination defects in composite structures, employing a Vibration-Based Damage Detection (VBDD) framework integrated with Wavelet Transform Mode Shape Curvature (WT-MSC). Beyond this specific focus, the study aims to conduct a comparative analysis of the efficacy of various detection methodologies, systematically assessing their operational viability and applicability while proposing novel analytical approaches for structural health monitoring.

This manuscript is organized as follows: Section 2 details the main concepts used in the development of this method; Section 3 demonstrates the methodology and workflows applied for damage identification; Section 4 presents the results obtained; and Section 5 summarizes the main conclusions of the study.

## 2. Theoretical framework

### 2.1. Vibration-based methods

In essence, Vibration-Based Damage Detection (VBDD) relies on analyzing the dynamic response of a structure, such as its natural frequencies, mode shapes, or Operational Deflection Shapes (ODSs), under excitation. While it is practically possible to estimate both natural frequencies and mode shapes in a single experimental step, this study deliberately adopts a two-phase methodological approach. First, a traditional point-based contact method (impact hammer and accelerometer) is used to acquire the Frequency Response Function (FRF), establishing a highly reliable "ground truth" baseline of the fundamental natural frequencies. Subsequently, a continuous shaker excitation is employed in conjunction with Digital Image Correlation (DIC) to extract the full-field ODSs at those specific frequencies. This separation ensures that the non-contact optical measurements are robustly validated against established contact techniques. ODSs, defined as the vibration pattern of a structure at a specific frequency [10], provide a global visual interpretation of the vibration data, enabling the localized identification of structural anomalies. ODSs are defined as the vibration pattern of a structure at a specific frequency [13]. Generally, the use of ODS for damage detection provides a visual interpretation of vibration data, enabling the identification of anomalies and structural failures. In this study, ODSs were obtained using Digital Image Correlation (DIC) rather than traditional numerical methods, a process that will be detailed in subsequent sections.

Damage detection is fundamentally based on the principle that the presence of damage causes a local reduction in stiffness. This reduction affects the vibration modes, creating irregularities at the damage location. This approach is often referred to as the Gapped Smoothing Technique [14,15]. To facilitate the visual analysis of these damage-induced differences, multiple derivatives are applied to the numerical data to highlight discrepancies. It is important to note that, theoretically, these differences exist in the raw vibration mode data; however, their amplitude is often too small for simple visual detection.

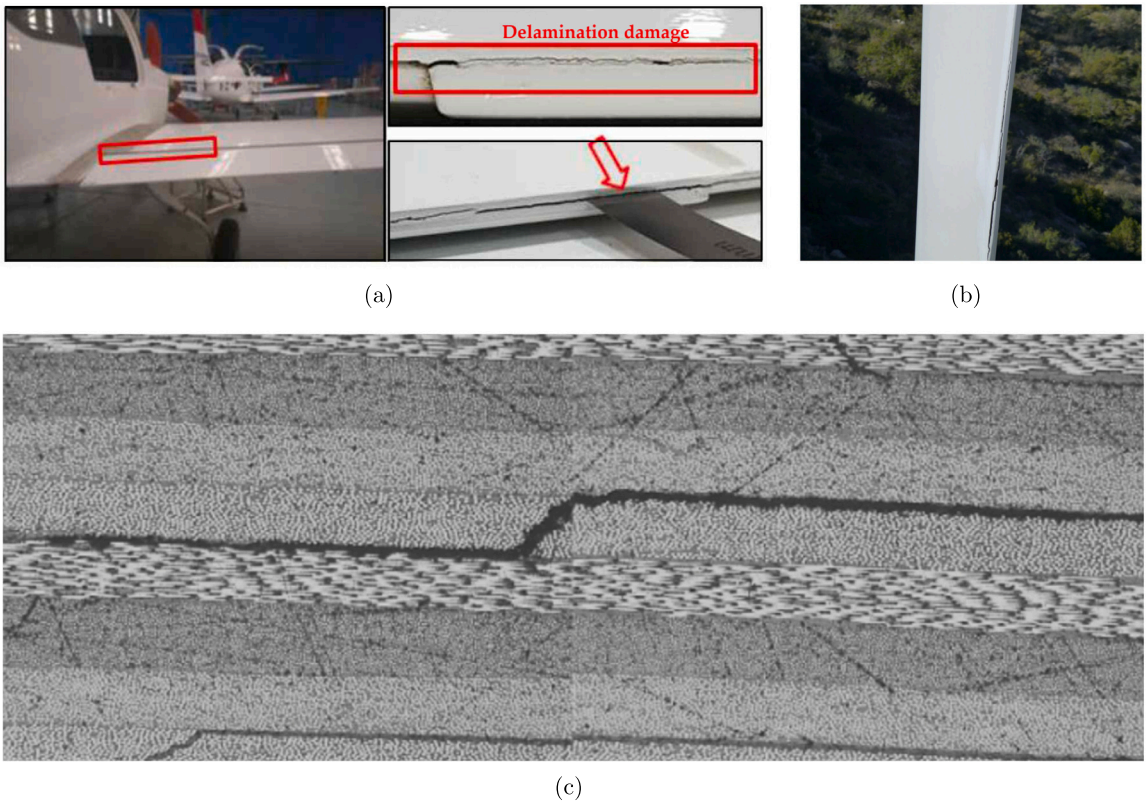



**Fig. 1.** (a) Skin-debonding damage in aircraft tail wing structure [10], damage on a wind turbine blade [11] and (c) a microscopic view of delamination ([12]).

Consequently, the Mode Shape Curvature (MSC) method [16] is employed, which utilizes the second numerical derivative of the vibration mode to maximize the difference caused by stiffness variations. When considering the MSC as a continuous surface, irregularities indicate the location of the damage. However, a significant drawback is that amplifying stiffness differences also amplifies measurement noise.

Another effective method is Wavelet Analysis [17]. This technique is capable of evidencing damage without excessive noise amplification, as it treats mode shape data as a signal in the spatial domain rather than the frequency domain. A further advantage over the MSC method is the ability of wavelets to support multi-scale analysis, allowing for the interrogation of hidden damage at various levels of detail.

#### 2.1.1. Mode difference approaches

In addition to the aforementioned standard methods, this study investigates three distinct approaches based on MSC theory [4]:

1. **Mode Shape Difference:** Comparing a plate with hidden damage against a pristine (undamaged) plate to confirm differences caused by stiffness alteration.
2. **Curvature Difference:** Calculating the curvature of both the damaged and pristine plates separately, then subtracting them to isolate the damage signal from the noise.
3. **Curvature of the Difference:** Subtracting the raw mode shapes first, and then applying multiple derivatives to the residual difference to highlight the damage.

#### 2.1.2. Modal curvature method (MSC)

The Modal Curvature Method (MSC) [16] utilizes the second derivative of the vibration mode to highlight hidden damages. This is based on the relationship between curvature and the flexural stiffness of a structure's cross-section, defined as:

$$\Phi'' = \frac{M}{EI} \tag{1}$$

where $\Phi''$ is the curvature, $M$ is the bending moment, $E$ is the Young's modulus, and $I$ is the area moment of inertia.

The fundamental principle of MSC is that a defect reduces the flexural stiffness ($EI$) at the damage location, leading to a local increase in curvature. The most common method for deriving modal curvature is the Central Difference Method:

$$\Phi''(i) \approx \frac{\Phi_{i+1} + \Phi_{i-1} - 2\Phi_i}{h^2} \tag{2}$$

where $\Phi$ refers to the modal displacement and $h$ is the sensor/point spacing.

For plate-like structures, the modal curvature can be expressed as the Laplacian operator of the mode shape:

$$\Phi''(i,j) = \nabla^2\Phi(i,j) \approx \frac{\Phi_{i+1,j} + \Phi_{i-1,j} - 2\Phi_{i,j}}{h_i^2} + \frac{\Phi_{i,j+1} + \Phi_{i,j-1} - 2\Phi_{i,j}}{h_j^2} \tag{3}$$

where $h_i$ and $h_j$ represent the grid spacing in the $x$ and $y$ directions, respectively.

It is important to acknowledge that the direct interpretation of $\nabla^2\Phi$ as a damage indicator is primarily formulated for isotropic plate theories, where curvature is inversely proportional to a uniform flexural rigidity. In composite laminates, inherent material anisotropy and potential bend–twist coupling introduce natural directional variations in the baseline curvature field, complicating direct damage identification. However, damage mechanisms such as delamination introduce highly localized, high-spatial-frequency singularities in the stiffness matrix. While the laminate's anisotropy dictates the global, low-frequency curvature topology, the defect creates an abrupt local discontinuity. Consequently, raw curvature maps often require advanced signal processing, such as the Wavelet Transform explored in subsequent sections, which effectively acts as a spatial filter to separate the global anisotropic baseline response from the high-frequency local anomalies induced by the damage.

It is crucial to emphasize that curvature analysis depends on numerical differentiation and is consequently highly susceptible to errors, particularly when the input data contains significant noise.

### 2.2. Noise reduction methods

Given that the derivation process amplifies noise levels in the data [18], which tends to mask results and obscure the true position of damage, mathematical smoothing and noise reduction functions are employed. This section describes the methods applied in this work.

#### 2.2.1. Gaussian filter

To reduce noise practically and effectively, a Gaussian filter was applied to the vibration modes $w(x_i, y_i)$ [19]. A standard 2D Gaussian binary function was used to construct the filter, defined by:

$$f(x,y) = \frac{1}{2\pi\sigma^2}\, e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{4}$$

where $\sigma$ is the standard deviation of the Gaussian distribution. Consequently, the multiple derivatives are applied to the vibration mode only after it has been processed by this filter.

Additionally, a Gaussian filter (scipy.ndimage) with a standard deviation of $\sigma = 2$ was applied to the displacement data. This specific value was empirically selected because it provided the optimal balance between effectively smoothing the high-frequency measurement noise inherent to the optical acquisition and preserving the localized spatial discontinuities (singularities) induced by the internal delaminations.

#### 2.2.2. Continuous wavelet transform

A Wavelet Transform (WT) is a mathematical tool that decomposes a signal into different scales and positions. Its most common application is the analysis of localized and intermittent oscillations. In the context of VBDD, the WT demonstrates a high capacity for denoising dynamic data while extracting specific features through multi-scale analysis [20].

As suggested by Wang et al. (2024), by projecting the image onto a 2D wavelet in the spatial domain (considering $x$, $y$, and a scale parameter), it is possible to create a family of wavelets $\psi_{s,\mathbf{b},\theta}(\mathbf{x})$ with scale, translation, and rotation indices, defined as:

$$\psi_{s,\mathbf{b},\theta}(\mathbf{x}) = \frac{1}{s}\,\psi\left[\mathbf{r}_\theta^{-1}\left(\frac{\mathbf{x}-\mathbf{b}}{s}\right)\right] \tag{5}$$

where $s$ is the scale parameter, $\mathbf{b} \in \mathbb{R}^2 = (b_1, b_2)^{\mathrm{T}}$ is the 2D translation parameter, $\mathbf{x} = (x, y)^{\mathrm{T}}$ is the 2D position vector, and $\mathbf{r}_\theta$ is the rotation matrix with angle $\theta$, given by:

$$\mathbf{r}_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{6}$$

where $0 \le \theta < 2\pi$.

The 2D function $\psi$ must satisfy the admissibility condition:

$$C_\psi = (2\pi)^2 \int_{\mathbb{R}^2} \frac{|\hat{\psi}(\mathbf{k})|^2}{|\mathbf{k}|^2}\, d^2\mathbf{k} < \infty \tag{7}$$

where

$$\hat{\psi}(\mathbf{k}) = \int_{-\infty}^{\infty} \psi(\mathbf{x})\, e^{-i\mathbf{k}\cdot\mathbf{x}}\, d^2\mathbf{x} \tag{8}$$

Here, $\hat{\psi}$ is the Fourier transform of $\psi$, $\mathbf{k} = (k_1, k_2)^{\mathrm{T}}$ is the wavenumber vector, $|\mathbf{k}| = \sqrt{k_1^2 + k_2^2}$, and $\mathbf{k}\cdot\mathbf{x} = k_1 x + k_2 y$. The admissibility condition implies that the wavelet must have a zero mean:

$$\hat{\psi}(0) = 0 \quad \text{or} \quad \int_{-\infty}^{\infty} \psi(\mathbf{x})\, d^2\mathbf{x} = 0 \tag{9}$$

This condition determines the transform's ability to detect irregularities. A wavelet function is said to have $n$ vanishing moments when:

$$\int_{\mathbb{R}^2} x^p y^q \psi(\mathbf{x})\, d^2\mathbf{x} = 0, \quad 1 \le p+q \le n \tag{10}$$

The 2D Continuous Wavelet Transform (CWT) of $f(\mathbf{x})$ can be expressed as:

$$W_\psi^f(s,\mathbf{b},\theta) = \frac{1}{s}\iint f(\mathbf{x})\,\psi^*\left[\mathbf{r}_\theta^{-1}\left(\frac{\mathbf{x}-\mathbf{b}}{s}\right)\right] d^2\mathbf{x} \tag{11}$$

where $f(\mathbf{x}) = f(x,y) \in L^2(\mathbb{R}^2)$ represents the 2D image data. Thus, it can be stated that the WT $W_\psi^f(s,\mathbf{b},\theta)$ is orthogonal to polynomial signals of degree up to $n-1$.

### 2.3. Use of DIC in VBDD

Digital Image Correlation (DIC) operates by capturing images of targets prepared with a stochastic pattern of random points ("speckles") applied to the surface. By comparing successive images of these points on a vibrating surface against an original, undeformed reference image, it is possible to track subsets of points and compute surface deformations across the entire field of view [21–23].

Through this method, the integration of DIC with high-speed cameras (High-Speed DIC) enables global, non-contact vibration measurements in the micrometer range. Generally, DIC is categorized into two types: Two-Dimensional DIC (2D-DIC) and Three-Dimensional DIC (3D-DIC).

2D-DIC requires only a single camera but is limited to measuring in-plane deformations ($x$ and $y$ directions) on planar structures. Conversely, 3D-DIC captures both in-plane and out-of-plane components ($x$, $y$, and $z$ directions), but relies on the analysis of image pairs captured simultaneously by a calibrated stereo system. This stereo system consists of two synchronized cameras recording a common Region of Interest (ROI) from specific viewing angles.

3D-DIC is considered superior for vibration measurements due to its ability to capture out-of-plane displacements, the magnitude of which is typically much larger than in-plane displacements for flexural vibration modes. However, 3D-DIC requires rigorous setup to ensure optimal system performance, as errors can arise from camera desynchronization or shifts in relative camera positions post-calibration.

Additionally, the DIC algorithm requires significantly higher processing time for 3D applications, as corresponding point pairs in the stereo images must be identified via triangulation before being correlated with the reference image pair. Nevertheless, the result of a well-executed calibration and measurement is a comprehensive representation of surface points in 3D space over time, making it uniquely suited for full-field vibration analysis.

In the context of mode shape data from DIC, the performance of the Curvature of Difference method was quantified using the following statistical indicators:

1. **Modal Assurance Criterion (MAC):** Measures the global correlation between the pristine ($\phi_{ref}$) and damaged ($\phi_{dmg}$) mode shapes. A value close to 1 indicates high similarity, implying that damage has not significantly altered the global modal topology.

$$MAC = \frac{|\phi_{ref}^T \phi_{dmg}|^2}{(\phi_{ref}^T \phi_{ref})(\phi_{dmg}^T \phi_{dmg})} \tag{12}$$

2. **Signal-to-Noise Ratio (SNR):** Evaluates the contrast between the damage signature and the background noise. It is defined as the difference between the mean damage index in the defective region ($\mu_{dmg}$) and the healthy region ($\mu_{healthy}$), normalized by the standard deviation of the healthy region ($\sigma_{healthy}$):

$$SNR = \frac{\mu_{dmg} - \mu_{healthy}}{\sigma_{healthy}} \tag{13}$$

3. **Receiver Operating Characteristic (ROC) and AUC:** The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate at various threshold settings. The Area Under the Curve (AUC) provides an aggregate measure of performance across all possible classification thresholds. An AUC of 0.5 represents a random guess, while 1.0 represents perfect detection.

**Table 1**
Geometric parameters and coordinates of the induced delaminations (Plate 1).

| Defect ID | Dimensions (mm) | Dimensions (in) | Position ($Y$) | Position ($X$) |
|---|---|---|---|---|
| D1 | 25.40 × 25.40 | 1.00 × 1.00 | Top (800 mm) | 187.3 mm |
| D2 | 19.05 × 19.05 | 0.75 × 0.75 | Top (800 mm) | 385.8 mm |
| D3 | 12.70 × 12.70 | 0.50 × 0.50 | Top (800 mm) | 579.5 mm |
| D4 | 6.35 × 6.35 | 0.25 × 0.25 | Top (800 mm) | 776.3 mm |
| D5 | 25.40 × 25.40 | 1.00 × 1.00 | Bottom (200 mm) | 187.3 mm |
| D6 | 19.05 × 19.05 | 0.75 × 0.75 | Bottom (200 mm) | 385.8 mm |
| D7 | 12.70 × 12.70 | 0.50 × 0.50 | Bottom (200 mm) | 579.5 mm |
| D8 | 6.35 × 6.35 | 0.25 × 0.25 | Bottom (200 mm) | 776.3 mm |

4. **Classification Metrics:** Using a fixed global threshold ($T = 0.3$) on the normalized damage index, the F1-Score (harmonic mean of Precision and Recall) was calculated to assess the localization accuracy. This specific threshold value was empirically tuned based on the preliminary ROC curve analysis to provide a standardized benchmark across all modes. It represents an empirical compromise aimed at filtering out low-level background noise while preserving the high-magnitude peaks (the top 30% of the normalized wavelet coefficients) corresponding to the structural singularities. However, because the Signal-to-Noise Ratio (SNR) is highly frequency-dependent, applying a single fixed threshold constitutes a conservative limitation of this quantitative evaluation. While sufficient for a baseline comparison, future fully automated SHM implementations would significantly benefit from frequency-dependent or adaptive statistical thresholding algorithms to account for the varying noise floor across different modal orders:

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \tag{14}$$

## 3. Methodology

### 3.1. Materials and specimens

The experimental campaign utilized two Carbon Fiber Reinforced Polymer (CFRP) laminate plates with nominal dimensions of 1000 mm× 1000 mm and a thickness of approximately 3 mm. Both plates were manufactured using an aerospace-grade layup sequence.

The first specimen (Plate 2) serves as the pristine baseline control, free of any intentional structural defects. The second specimen (Plate 1) contains eight artificial delaminations introduced during the manufacturing process to simulate internal damage. These delaminations were created by inserting thin Polytetrafluoroethylene (PTFE/Teflon) films between the composite plies, a standard method for preventing interlayer bonding without significantly altering the local thickness or mass.

The induced damage comprises eight square-shaped delaminations of varying dimensions, strategically distributed to evaluate the detection sensitivity of the proposed method across different defect severities. These defects were simulated using non-adhesive PTFE (Teflon) inserts placed at specific interlaminar interfaces during the layup process.

Table 1 details the geometric specifications and spatial arrangement of the defects. The delaminations are arranged in two parallel rows (Top and Bottom), with dimensions ranging from 25.4 mm (1.00 in) to 6.35 mm (0.25 in), representing a damage-to-area ratio spectrum critical for validation.

These defects are spatially distributed in two parallel rows across the plate surface and are located at different ply interfaces (depths) to represent varying degrees of severity and detectability difficulty. Fig. 2 illustrates the technical schematic of the damaged plate, detailing the positioning and geometric specifications of the inserts.

To verify the experimental conditions and the quality of the optical preparation, Fig. 3 illustrates the specimens in the testing rig. Figures show the global setup for the damaged and pristine plates, respectively. To ensure high-contrast tracking for the DIC algorithm, a stochastic speckle pattern was applied to the surface. Additionally, the locations of the induced delaminations were marked on the surface for visual reference during the initial inspection.

### 3.2. Experimental setup

To ensure high-fidelity data acquisition, a high-speed stereoscopic camera system from Dantec Dynamics was employed. The setup consisted of two cameras (DSM 16520 CXP-12 | 5MPx | 700fps | 4/3" sensor) equipped with 16 mm C-Mount lenses (VT-Lens, f=16mm, 12Mpx | MOD: 100 mm). The acquisition rate was set to 700 fps at maximum resolution. Illumination was provided by a pair of high-frequency red LED floodlights to prevent flickering. All equipment was leveled using a heavy-duty tripod and a laser level to ensure geometric alignment.

Both laminated composite plates were analyzed under free vibration condition. Plates 1 and 2, provided by Embraer, possess identical dimensions (1 m×1 m, 2.8 mm thickness). The distinguishing feature of Plate 1 is the presence of eight induced delaminations of varying sizes located at different ply depths. Both plates were analyzed in a free–free boundary condition, suspended by a set of nylon cables attached to springs to simulate an unsupported state.

Prior to testing, the camera system underwent a calibration procedure to capture a calibration panel composed of points with known distances in various orientations covering the entire Region of Interest (ROI). The panel, manufactured by Dantec Dynamics, features uniformly sized dots spaced 10 mm apart on a rigid substrate.

As indicated by Wang et al. [4], the objective of calibration is to determine the intrinsic parameters of each camera (e.g., distortion parameters, focal length) and the extrinsic parameters of the array (relative angles and baseline spacing). These parameters are essential for computing the 3D transformations from 2D image coordinates via stereo-triangulation. The entire system was interfaced with the VIC-3D software for acquisition and analysis.

### 3.3. Modal testing

To establish a structural baseline, the Frequency Response Function (FRF) of the plates was acquired. This step is fundamental, as the damage identification methodology relies heavily on the accurate extraction of vibration modes. An initial experimental modal analysis was conducted using an impact hammer to provide a broadband impulse excitation. The structural response was measured using a single accelerometer bonded to the plate surface, as shown in Fig. 4(a). This procedure facilitated the identification of the theoretical natural frequencies and mode shapes.

Subsequently, the time-domain acceleration data were transformed into the frequency domain using a Fast Fourier Transform (FFT). This process yielded the frequency response spectra for both CFRP plates, which served as the reference data for all subsequent post-processing and damage detection comparisons.

The instrumentation utilized for excitation and response measurement is detailed in Fig. 4.

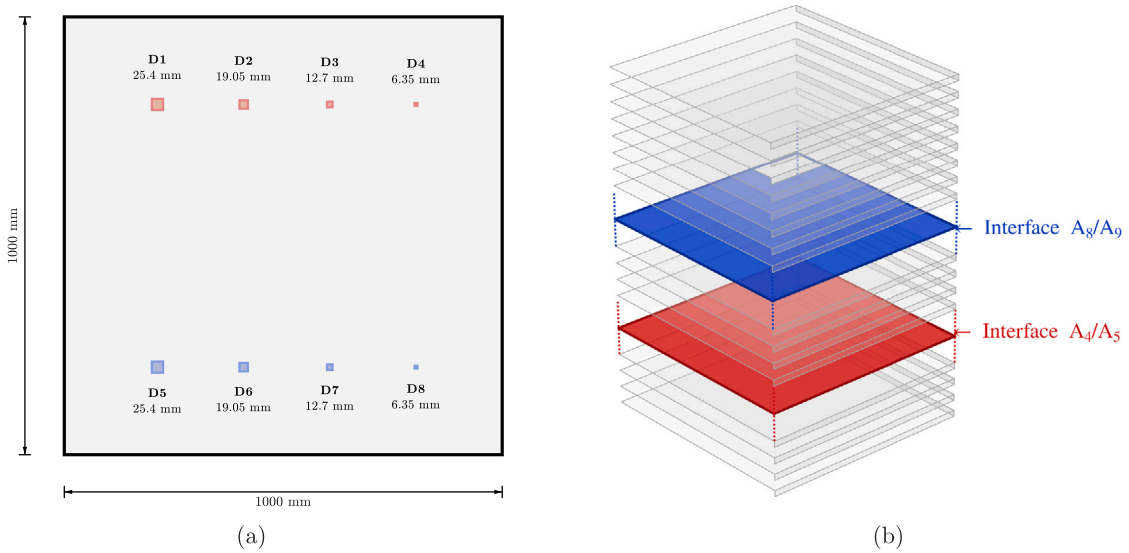



**Fig. 2.** Schematic representation of the CFRP composite specimen with induced delaminations. (a) Top view detailing the global in-plane dimensions (1000 × 1000 mm) and the spatial distribution of the eight square Teflon inserts, which range in size from 6.35 mm to 25.4 mm. (b) 3D exploded view illustrating the 16-ply stacking sequence and the specific through-thickness locations of the defects at interfaces A4/A5 (shallower) and A8/A9 (mid-plane).

### 3.4. Digital image correlation (DIC)

To validate the optical measurement approach, the modal analysis was replicated using Digital Image Correlation (DIC). This non-contact, full-field optical technique functions as a dense array of "virtual sensors", capable of measuring surface displacements over time with high spatial resolution.

For the initial analysis of the full-scale Plates 1 and 2, the structures were excited within a frequency range of 1–300 Hz using a linear sine sweep signal (chirp) with a duration of 1 s. The excitation was introduced via an electrodynamic shaker attached to the specimen extremity, as detailed in Fig. 4(c). The excitation was introduced via an electrodynamic shaker coupled to the lower extremity of the specimen. To ensure pure axial force transmission and minimize unwanted rotational constraints that could artificially stiffen the structure, the shaker was connected to the plate using a thin, flexible steel stinger.

Subsequently, a refined experiment was conducted on Plate 1, which was subdivided into two sections (upper and lower) for analysis. In this scenario, the frequency range was extended to 1–800 Hz. This adjustment was necessary to capture higher-order modes; however, achieving the higher frame rates required to satisfy the Nyquist–Shannon sampling theorem (sampling at least twice the highest frequency of interest) necessitated a reduction in the camera's active sensor area. Consequently, this phase involved a trade-off between the spatial field-of-view (FOV) and the temporal bandwidth.

To ensure effective correlation, the specimen surface must possess a non-repetitive, isotropic contrast pattern. Accordingly, the plates were prepared by applying a white matte base coat, followed by the application of a stochastic pattern of black speckles using textured rollers provided by Correlated Solutions. The complete experimental configuration, including the stereoscopic camera positioning and the optimized speckle pattern, is illustrated in Fig. 5.

Post-correlation, the time-domain displacement data were processed using the acquisition software's integrated Fast Fourier Transform (FFT) algorithm. This process extracted the natural frequencies and Operational Deflection Shapes (ODS). Finally, these datasets were exported to a Python environment for advanced post-processing and comparison with the accelerometer data.

Regarding experimental repeatability and signal variability, it is important to clarify that while the experimental campaign utilized a single continuous frequency sweep per setup, the extracted mode shapes are statistically robust. The high-speed optical acquisition (700 fps) generated a dense time–history of full-field displacements. The subsequent Fast Fourier Transform (FFT) inherently averages this dynamic response over the temporal window. This time-averaging process significantly reduces random experimental noise and temporal variability, ensuring that the extracted Operational Deflection Shapes (ODS) represent a stable and reliable realization of the structural dynamics.

### 3.5. Signal processing

#### 3.5.1. Wavelet transform - mode shape curvature (WT-MSC)

To mitigate the noise amplification caused by numerical differentiation of raw vibration modes, Wang et al. (2024) proposed the Wavelet Transform-Mode Shape Curvature (WT-MSC) method. This approach circumvents the need for direct second-order derivation of the mode shape by utilizing the second derivative of the Wavelet function itself.

For a modal displacement $f(x)$ with modal curvature $f''(x)$, the WT-MSC is defined by the convolution property:

$$f''(x) * \varphi_{s,a}(x) = f(x) * \varphi''_{s,a}(x) = f(x) * \psi_{s,a}(x) \tag{15}$$

where $\psi_{s,a}(x)$ is the second derivative of the scaling function $\varphi_{s,a}(x)$.

For a 2D plate, the "mother" wavelet is redefined as:

$$\psi_{s,a,b}(x,y) = \frac{1}{s}\,\psi\left(\frac{x-a}{s}, \frac{y-b}{s}\right) \tag{16}$$

Mirroring the 2D smoothing function:

$$\varphi_{s,a,b}(x,y) = \frac{1}{s}\,\varphi\left(\frac{x-a}{s}, \frac{y-b}{s}\right) \tag{17}$$

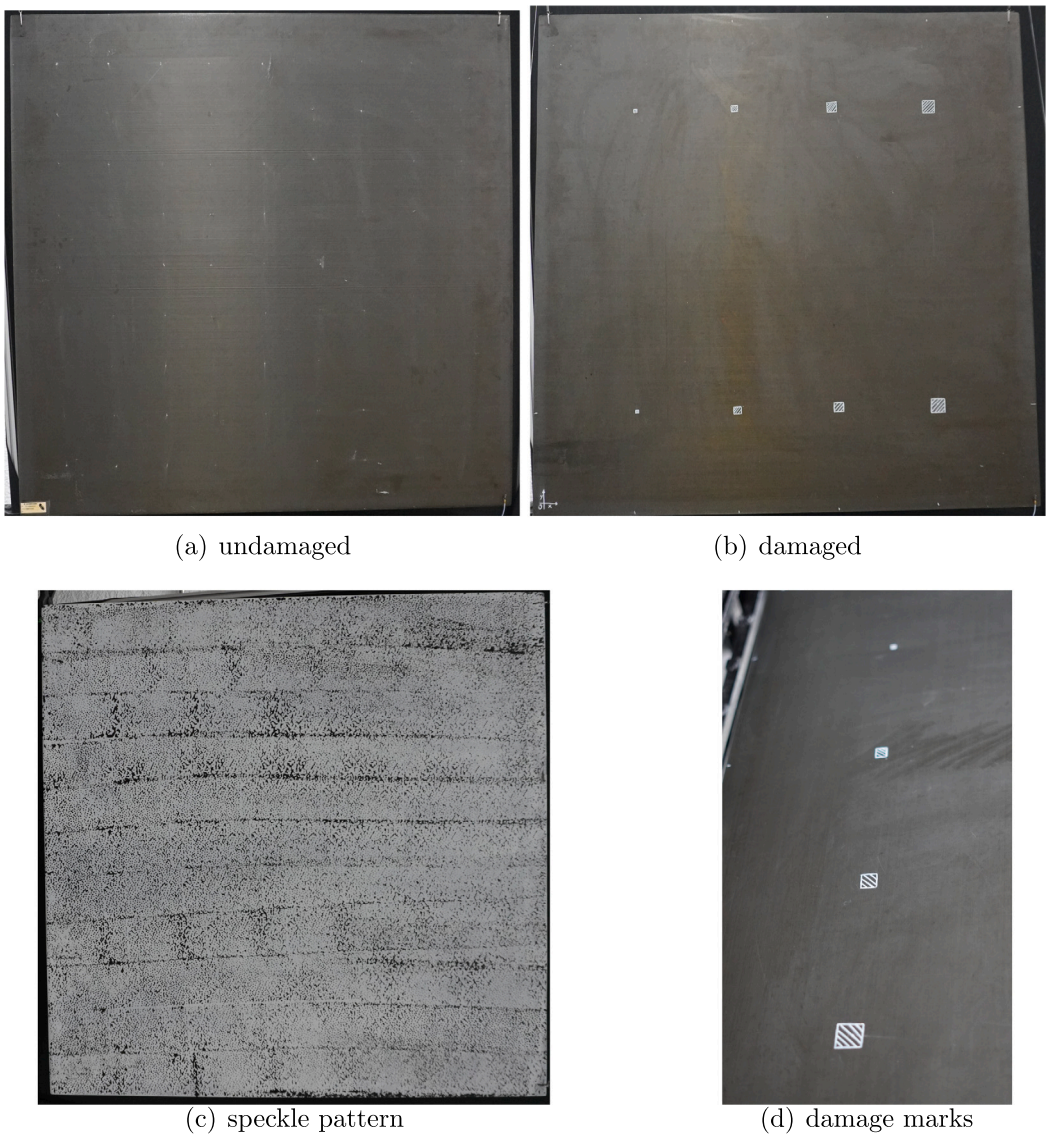
The wavelet coefficient is then given by:

$$W_f^{\psi}(s,a,b) = \iint_{\mathbb{R}^2} f(x,y)\,\psi^*_{s,a,b}(x,y)\,dx\,dy \tag{18}$$

$$= \frac{1}{s}\iint_{\mathbb{R}^2} f(x,y)\,\psi^*\left(\frac{x-a}{s}, \frac{y-b}{s}\right)\,dx\,dy \tag{19}$$

For a 2D mode shape $f^{(i)}(x,y)$ of mode $i$, with modal curvature represented by the Laplacian operator $\nabla^2 f^{(i)}(x,y)$, the WT-MSC relationship is:

$$W^{\psi}_{\psi f^{(i)}}(s,a,b) = \nabla^2 f^{(i)}(x,y) * \varphi_{s,a,b}(x,y)$$
$$= f^{(i)}(x,y) * \nabla^2\varphi_{s,a,b}(x,y) = f^{(i)}(x,y) * \psi_{s,a,b}(x,y) \tag{20}$$



(a) undamaged (b) damaged

(c) speckle pattern (d) damage marks

**Fig. 3.** Experimental configuration and surface preparation details of the CFRP specimens.

*3.5.2. Mexican hat wavelet*

In this study, the Mexican Hat wavelet was selected for the WT-MSC analysis. This wavelet is the negative normalized second derivative of a Gaussian function and possesses two vanishing moments. The primary advantage of this wavelet, as highlighted by Wang et al. (2024) [4], is its efficacy in detecting singularities at finer scales. The Mexican Hat function is defined as:

$$\psi(x,y) = \nabla^2\varphi(x,y) = \frac{1}{\pi\sigma^4}\left(1 - \frac{x^2+y^2}{2\sigma^2}\right) e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{21}$$

where $\sigma$ is the standard deviation and $\varphi(x,y)$ is the Gaussian function:

$$\varphi(x,y) = \frac{1}{2\pi\sigma^2}\, e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{22}$$

It is important to note regarding the general 2D wavelet formulation presented previously that the rotation parameter $\theta$ is effectively unnecessary when utilizing the Mexican Hat wavelet. As indicated by Equations, the function depends solely on the squared radial distance ($x^2 + y^2$), making it perfectly isotropic (radially symmetric). Consequently, rotating this specific wavelet by any angle $\theta$ yields an identical mathematical function. Because rotational invariance is inherently satisfied by the wavelet's structural symmetry, computing the transform across multiple orientation angles is mathematically redundant. Therefore, the analysis assumes $\theta = 0$ without any loss of directional sensitivity for identifying randomly oriented structural singularities.

*3.5.3. Data processing implementation*

Upon completion of the experiments, data were exported in .csv format and processed using Python scripts leveraging the pandas library. A crucial pre-processing step involved removing null values and regularizing the data grid. Normalization of the Cartesian axes and interpolation of displacement data were performed using the scipy.interpolate library.

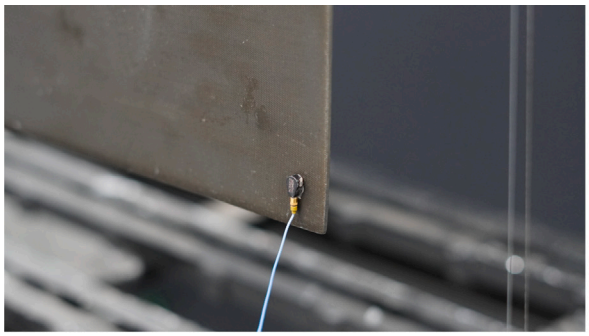
Normalization was applied according to:

$$x_{\text{norm}} = a \cdot \frac{x - \min(x)}{\max(x) - \min(x)} \tag{23}$$

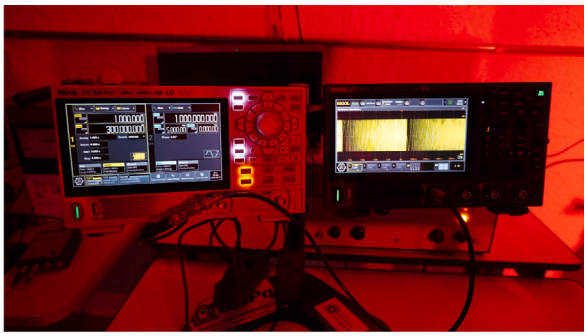
$$y_{\text{norm}} = b \cdot \frac{y - \min(y)}{\max(y) - \min(y)} \tag{24}$$

where $a$ and $b$ define the desired metric intervals for $x$ and $y$. Interpolation was achieved via Delaunay triangulation of points $(x,y)$, followed by surface approximation of the displacement $z$:

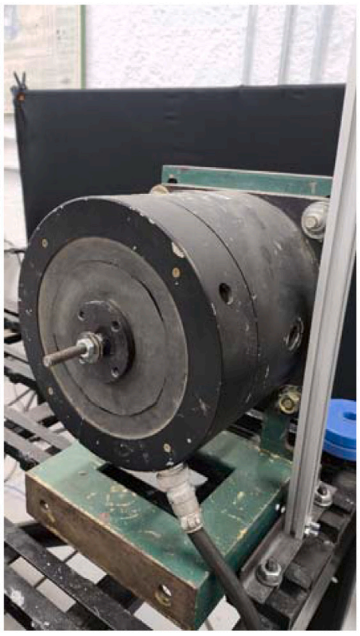
$$z(x,y) = \sum_{\substack{i+j+k=3 \\ i,j,k\geq 0}} a_{ijk}\, \lambda_1^i\, \lambda_2^j\, \lambda_3^k \tag{25}$$



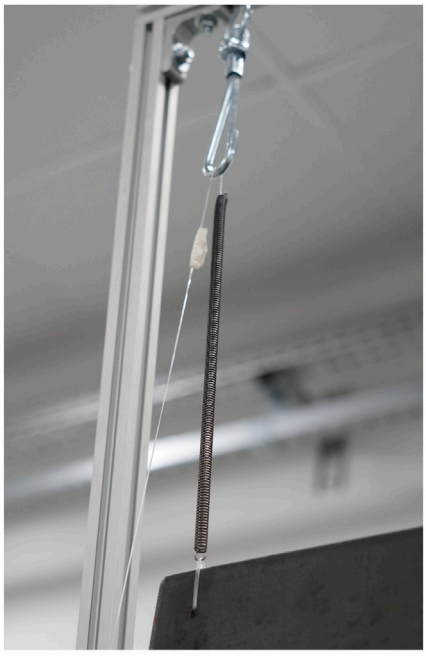
(a) Microaccelerometer attached to the plate surface.



(b) Function generator used for signal control.



(c) Electrodynamic shaker for continuous excitation.



(d) Elastic wires simulating free-free boundary conditions.

**Fig. 4.** Experimental instrumentation used for modal testing, excitation, and boundary condition simulation.

where $\lambda_{1,2,3}$ are the barycentric coordinates of point $(x, y)$ within the triangle, and $a_{i,j,k}$ are the control coefficients of the polynomial. This cubic interpolation over the unstructured Delaunay grid was computationally implemented using the Clough–Tocher scheme [24]. This method constructs a piecewise cubic Bézier polynomial that guarantees global $C^1$ continuity (smooth first derivatives) across all triangle boundaries. Establishing this continuous differentiability is a strict mathematical requirement for this study, as the subsequent Mode Shape Curvature (MSC) analysis relies on evaluating second-order spatial derivatives, which would improperly evaluate to zero if standard linear interpolation were employed.

Heatmaps were generated using matplotlib. Additionally, a Gaussian filter (scipy.ndimage) was applied to the displacement data to smooth high-frequency noise inherent to the experimental acquisition.

With the structured grid defined, baseline-dependent analysis was performed by computing the element-wise difference between the displacement fields of the pristine (Plate 2) and damaged (Plate 1) structures:

$$\phi_{\text{diff}} = \phi_{Sd} - \phi_{Cd} \tag{26}$$

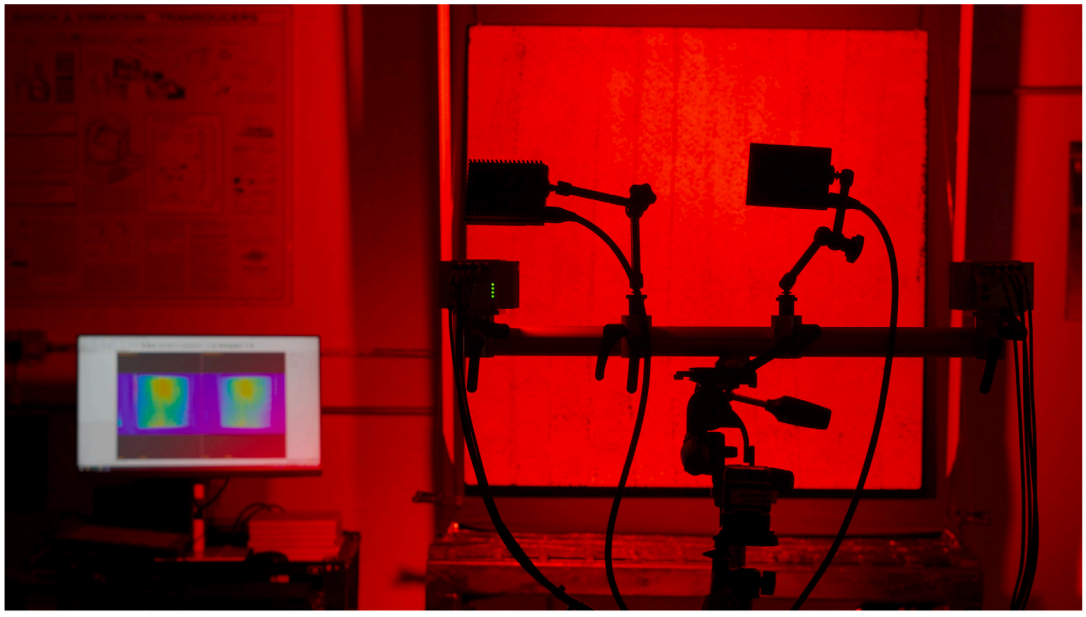
where $\phi_{Sd}$ and $\phi_{Cd}$ represent the mode shape data for the pristine (Sem dano) and damaged (Com dano) plates, respectively.

For curvature analysis, the gradient operator from scipy.ndimage was utilized to compute the second numerical derivative using the Central Finite Difference method:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h} \tag{27}$$

where $h$ is the spatial step size. This operation was applied to calculate the difference of curvatures, the curvature of differences, and the direct curvature of the damaged modes. Finally, as proposed, the Mexican Hat Wavelet Transform was applied via the pywt library directly onto the interpolated displacement data $z$, leveraging the convolution property derived in the previous section.

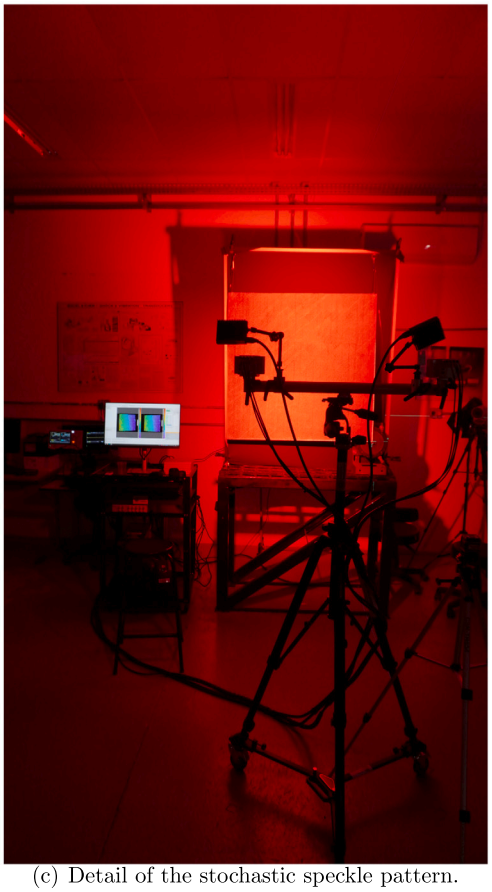
Regarding the implementation of the Continuous Wavelet Transform (CWT), it is crucial to address the selection of the wavelet scales. In this study, a single fixed scale was not adopted; instead, multiple scales were evaluated simultaneously to ensure optimal singularity detection. Specifically, a continuous scale vector ranging from 1 to 80 ($s \in [1, 80]$) was applied. The sensitivity of the damage maps is highly dependent on this scale parameter. Smaller scales (fine spatial resolution) are acutely sensitive to the abrupt, high-frequency spatial discontinuities caused by small-scale delaminations. Conversely, larger scales predominantly capture the low-frequency, global topology of the



(a) Stereoscopic camera array and lighting.



(b) System alignment and calibration.



(c) Detail of the stochastic speckle pattern.

**Fig. 5.** Experimental setup for the Digital Image Correlation (DIC) analysis, showing the camera positioning and surface speckle pattern.

operational deflection shapes. By evaluating a broad and continuous range of multiple scales, the resulting scalograms effectively separate the noise and global modal trends from the sharp singularities, mitigating the risk of missing damage due to an improperly tuned fixed scale.

To synthesize the experimental and analytical procedures described in the preceding sections, the complete methodological workflow is illustrated in Fig. 6. The process begins with the theoretical framework and experimental preparation, including camera calibration and speckle pattern application. Data acquisition is bifurcated into two phases: a baseline modal testing using an impact hammer and accelerometer to establish reference Frequency Response Functions (FRF), followed by the primary 3D-DIC acquisition using shaker excitation.

The acquired displacement data undergoes signal processing via Fast Fourier Transform (FFT) to extract frequencies and Operational Deflection Shapes (ODS). These datasets are then exported for advanced post-processing in Python, where they are subjected to interpolation and Gaussian filtering. Finally, the damage identification capabilities of three distinct methods—Mode Difference, Mode Shape Curvature (MSC), and Wavelet Transform Mode Shape Curvature (WT-MSC)—are evaluated and validated against the baseline FRF data.

To rigorously contextualize the reliability of the post-processing pipeline, it is essential to quantify the baseline displacement noise of the 3D-DIC system and understand its propagation. Given the 1 m×1 m macroscopic Field of View (FOV) and the 5-megapixel sensor resolution (2448 × 2048 pixels), the physical dimension of a single pixel is approximately 400 μm. Assuming a standard commercial DIC sub-pixel registration accuracy of 1/50th of a pixel, the baseline out-of-plane displacement uncertainty (noise floor) is conservatively estimated at 8 to 10 μm. This inherent measurement uncertainty propagates non-linearly through the analytical framework. Initially, the cubic interpolation acts as a spatial low-pass filter, slightly mitigating uncorrelated random noise. However, the subsequent numerical differentiation required for the Mode Shape Curvature (MSC) quadratically amplifies any residual high-frequency noise, dividing the spatial error by the square of the grid spacing ($h^2$). This catastrophic noise amplification is precisely what obscures the damage in standard curvature maps. Finally, the application of the Continuous Wavelet Transform (CWT) mitigates this degradation. Because wavelet convolution involves spatial integration, it functions as an advanced band-pass filter, suppressing the artificially amplified high-frequency artifacts while resonating with the coherent spatial singularities induced by the delaminations, thereby recovering the damage signature.

## 4. Results and discussion

This section presents the results obtained from the Digital Image Correlation (DIC) analysis and evaluates the effectiveness of the applied post-processing algorithms. To systematically address the challenges of structural health monitoring, the following analysis is explicitly divided into two distinct operational categories: Baseline-Dependent Analysis and Baseline-Free Analysis.

**Baseline-Dependent Methods** (Sections 4.3 to 4.5) rely on the direct mathematical comparison between the pristine (Plate 2) and damaged (Plate 1) states. While useful for academic validation, the primary limitation of these methods is their absolute requirement for a perfect, undamaged reference structure, which is rarely available for in-service aerostructures. Furthermore, they are highly susceptible to false positives caused by slight variations in boundary conditions, environmental noise, or sensor misalignment between separate measurement campaigns.

Conversely, **Baseline-Free Methods** (Sections 4.6 to 4.8) analyze solely the operational deflection shapes of the damaged structure, representing a much more realistic SHM scenario. The primary limitation of standard baseline-free techniques (such as raw Mode Shape Curvature) is the severe amplification of experimental noise during numerical differentiation, which can easily obscure the signature of small-scale defects. This specific limitation justifies the introduction of the WT-MSC approach (Section 4.8), which overcomes these barriers by utilizing wavelet convolution to suppress noise and isolate damage-induced singularities without ever requiring a pristine reference.

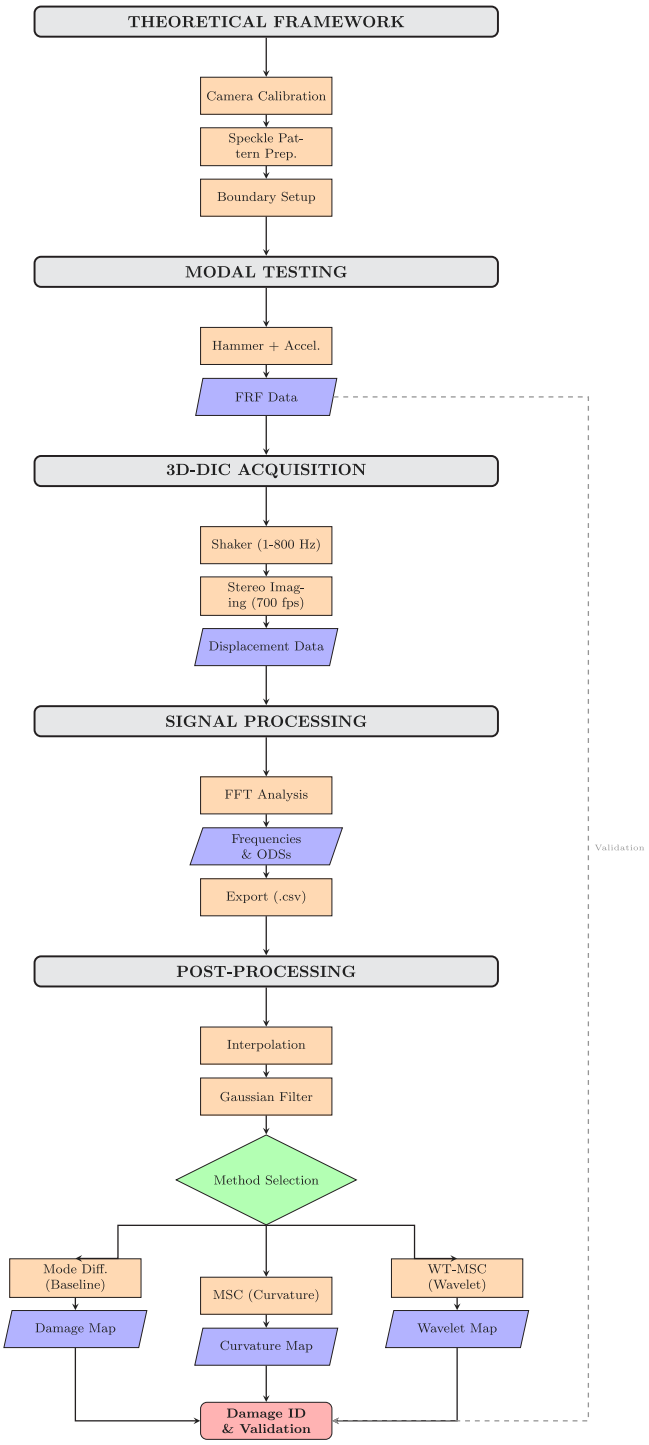
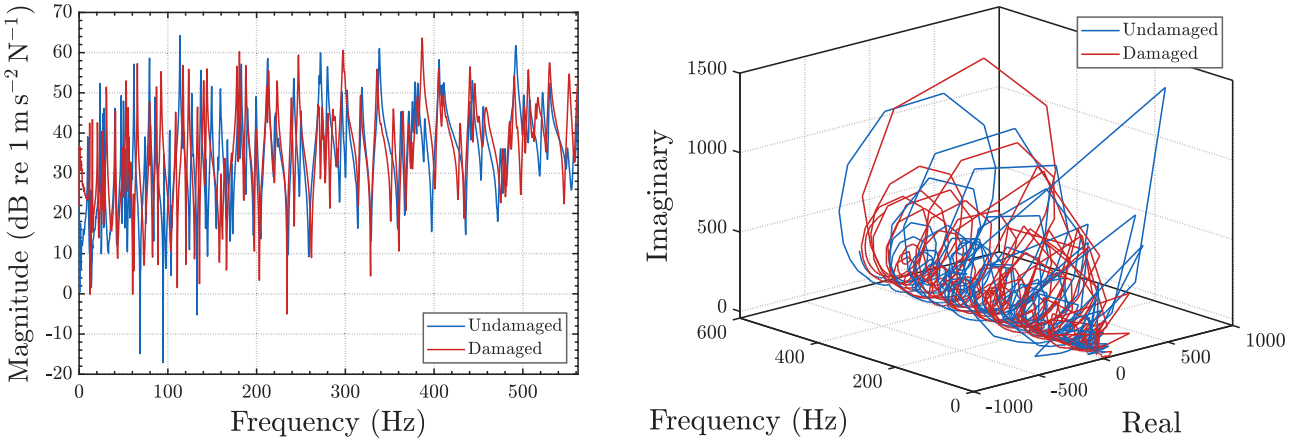



**Fig. 6.** Flowchart of the proposed methodology, detailing the stages from experimental setup and data acquisition (Modal Testing and 3D-DIC) to signal processing and damage identification algorithms.




(a) FRF Magnitude Comparison (dB). (b) 3D Nyquist Plot (Real vs. Imaginary vs. Freq).

**Fig. 7.** Comparison of experimental Frequency Response Functions (FRF) for Undamaged (Blue) and Damaged (Red) plates. (a) The magnitude plot highlights the frequency shifts due to stiffness reduction. (b) The 3D plot visualizes the complex response deviation across the frequency spectrum.

### 4.1. Free vibration analysis

The experimental modal analysis was initiated using an impact hammer and a single-point accelerometer to establish a robust baseline for the structural dynamics of the composite specimens. This conventional contact method provided high-coherence Frequency Response Functions (FRF), serving as the ground truth for validating the global, full-field data subsequently acquired via Digital Image Correlation (DIC).

Fig. 7 presents the instance FRFs obtained for the pristine (Plate 1) and damaged (Plate 2) structures. The response spectra clearly identify the first 12 fundamental vibration modes within the 0–500 Hz bandwidth.

To visualize the global impact of the delaminations, Fig. 7(a) overlays the magnitude of the FRF for both states. A discernible shift in the natural frequencies is observed, particularly in higher-order modes, where the stiffness reduction caused by the defects results in a leftward shift of the resonant peaks. This phenomenon is further elucidated in the 3D Nyquist plot shown in Fig. 7(b), which plots the Real versus Imaginary components against frequency. The deviation between the blue (Undamaged) and red (Damaged) trajectories confirms that the damage has altered the global mass-stiffness distribution, although the overall modal topology remains intact.

These frequency shifts, while measurable, are relatively small (often $< 1-5\%$), demonstrating the limitations of relying solely on global frequency changes for damage detection in complex composite structures. This observation justifies the proposed methodology's focus on spatially resolved curvature and wavelet metrics to localize defects that are otherwise "hidden" in the global response.

### 4.2. Mode shapes via DIC

To characterize the full-field dynamic response and investigate the spatial distribution of structural damage, the composite plates were analyzed using High-Speed 3D-DIC. This phase of the methodology follows the fundamental modal identification established in the previous section, expanding the analysis from discrete point-measurements to high-resolution surface mapping. The specimens were subjected to a broadband excitation using a continuous linear sine sweep signal (chirp) driven by an electrodynamic shaker, ensuring the simultaneous stimulation of multiple resonance frequencies.

For each discretized point within the 3D-DIC surface mesh, the high-speed optical acquisition generates a time–history of out-of-plane displacement, $w(x, y, t)$. A Fast Fourier Transform (FFT) is subsequently applied to these temporal signals, converting the data into the frequency domain to reconstruct the Operational Deflection Shapes (ODS) at the identified resonant peaks. To ensure optimal visual clarity, Fig. 8 presents a representative subset of these identified mode shapes, comparing the Undamaged (U) and Damaged (D) plates. The comprehensive dataset, containing the full grid of all 18 identified ODSs, is provided in its entirety in the Supplementary Material accompanying this article.

A qualitative inspection of these results reveals two critical observations. First, the global topological consistency is preserved; the fundamental bending and torsional mode shapes remain identifiable in both plates, indicating that the small-scale delaminations do not induce a catastrophic alteration of the global structural topology. Second, distinct frequency shifts are evident despite this visual similarity. For instance, the 4th mode shifts from 24.64 Hz in the pristine state to 22.82 Hz in the damaged state, a reduction directly attributable to the localized stiffness loss induced by the delaminations. These frequency shifts, validated by the global ODS analysis, serve as the primary indicator of damage presence, providing the necessary foundation for the subsequent application of the localized WT-MSC damage detection algorithms.

### 4.3. Mode shape difference analysis

The initial damage assessment was performed by calculating the algebraic difference between the mode shapes of the pristine (Plate 1) and damaged (Plate 2) structures. Fig. 9 illustrates the resulting difference maps for the identified resonant frequencies. High-intensity regions in these maps indicate areas where the local stiffness variation caused by damage led to significant deviations in the mode shape curvature or displacement field.

However, it is critical to note that these deviations predominantly manifest as global distortions rather than localized singularities. The difference maps tend to follow the macroscopic topology of the Operational Deflection Shapes (ODS), where the largest differences correspond to the areas of maximum displacement (antinodes) rather than the specific coordinates of the defects. While this method successfully confirms the existence of structural anomalies—validating the global stiffness reduction—it lacks the spatial resolution required to pinpoint the boundaries of small-scale delaminations. This limitation underscores the necessity for derivative-based metrics, such as Mode Shape Curvature (MSC) and Wavelet Transforms, to isolate local discontinuities from global modal behavior.

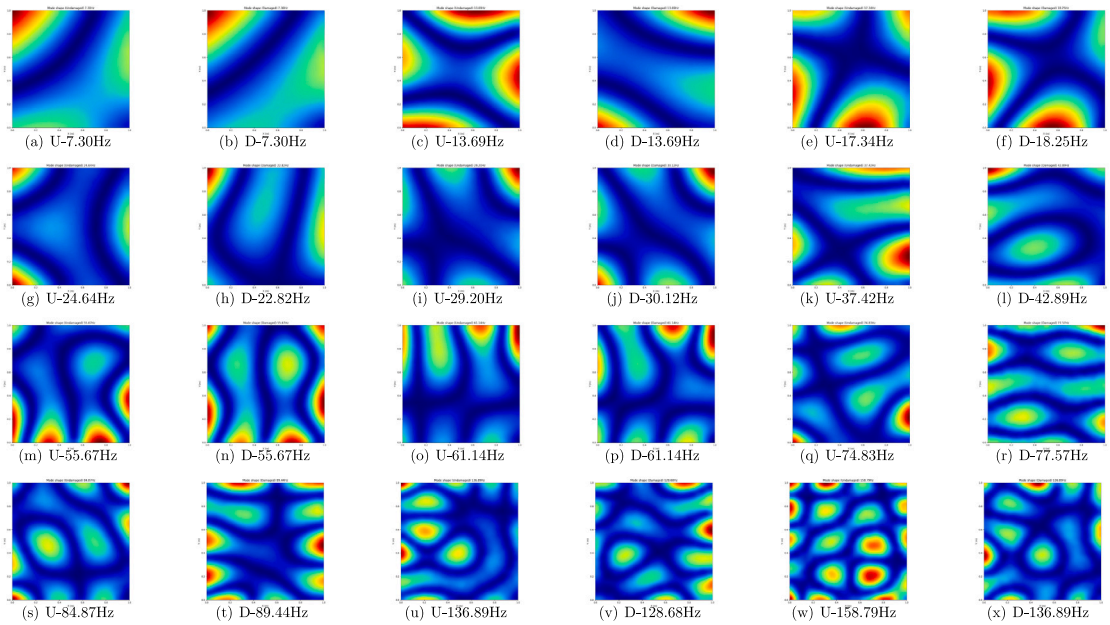



**Fig. 8.** Full-field visualization of the identified vibration modes. Each pair compares the Undamaged (U) and Damaged (D) condition for the identified natural frequencies. The layout proceeds in ascending frequency order from top-left to bottom-right.

### *4.4. Mode shape curvature (MSC) analysis*

Subsequent to the displacement analysis, the Mode Shape Curvature (MSC) method was applied to enhance damage visibility. Curvature is theoretically more sensitive to local stiffness reductions than displacement. Fig. 10 displays the curvature difference maps. The central finite difference approximation was used to compute the second derivative of the mode shapes, highlighting discontinuities that correspond to potential damage locations.

As a result, while the MSC method successfully highlights localized variations that were smoothed out in the displacement maps, the structural anomalies are partially obscured by the background noise floor. For example, in higher-order modes such as $\Delta$112.25 vs 109.51 Hz (Fig. 10m), the damage signatures are competing with differentiation artifacts. This observation highlights the limitations of using raw finite differences on experimental data and suggests the need for robust signal processing techniques, such as wavelet analysis, to improve the signal-to-noise ratio.

### *4.5. Curvature of difference analysis*

In addition to the Mode Shape Curvature difference, the Curvature of Difference method was evaluated. While conceptually similar, this approach first calculates the difference between the displacement fields and subsequently computes the curvature of that residual field. Fig. 11 presents the results, which often serve to isolate damage-induced singularities from global mode shape variations.

As observed in plots such as 107.69 vs. 109.51 Hz (Fig. 11l), the resulting maps display distinct localized "ripples" or singularities that were previously masked by the global curvature topology. This technique successfully isolates damage-induced stiffness changes from the global vibration characteristics, although it remains sensitive to the quality of the initial interpolation and grid regularization.

### *4.6. Curvature analysis of the damaged plate*

To evaluate the reference-free damage detection capabilities of the methodology, the Mode Shape Curvature (MSC) was computed solely for the damaged structure (Plate 2). Unlike the difference methods, this approach does not rely on a baseline measurement from the pristine structure. Fig. 12 presents the curvature fields. In these plots, localized spikes or irregularities in the curvature surface are potential indicators of damage, as the reduction in flexural stiffness causes a sharp increase in the local curvature.

### *4.7. Continuous Wavelet Transform (CWT) analysis*

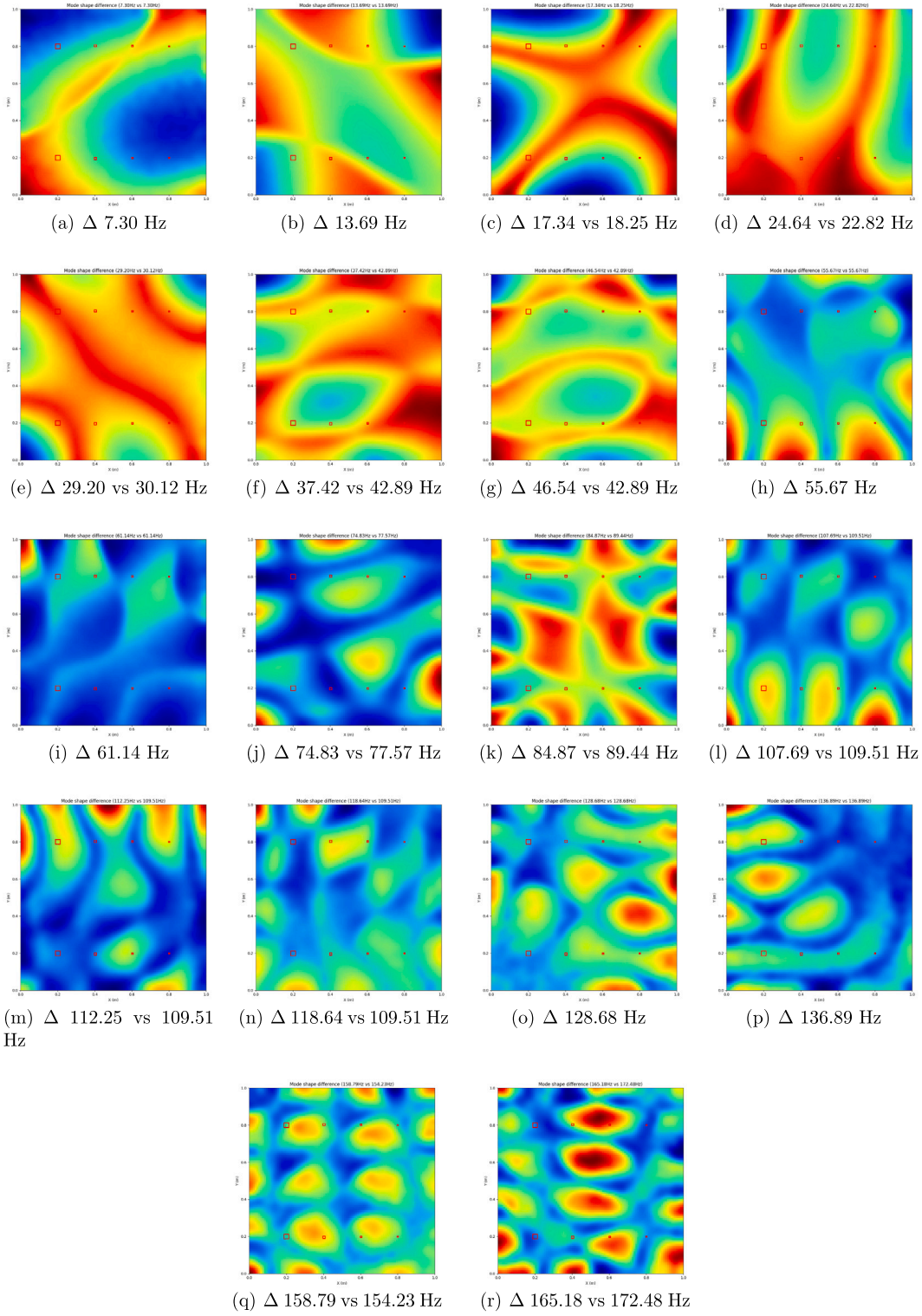
Finally, a signal processing approach based on the Continuous Wavelet Transform (CWT) was applied to the damaged mode shapes. Unlike the finite difference method used for curvature, the CWT provides a multi-resolution analysis that is exceptionally effective at detecting singularities in spatial signals.

The Mexican Hat wavelet (mexh) was selected as the mother wavelet due to its mathematical similarity to the second derivative operator (Laplacian), making it intrinsically sensitive to changes in curvature. Fig. 13 illustrates the spatial CWT coefficient maps. In these plots, the wavelet coefficients act as a damage index: sharp peaks or localized disturbances in the coefficient field indicate the presence of a structural discontinuity (damage) without requiring a healthy baseline for subtraction.

### *4.8. Wavelet Transform of Mode Shape Curvature (WT-MSC)*

Building directly upon the limitations of the raw Mode Shape Curvature (MSC) identified in previous section, specifically its susceptibility to catastrophic noise amplification, this section evaluates the efficacy of the proposed WT-MSC methodology. Because the fundamental challenge of differentiating experimental optical data has already been established, the following analysis focuses exclusively on demonstrating how the 2D-CWT spatial filtering overcomes these specific artifacts to successfully isolate the damage-induced singularities.

The application of the Wavelet Transform Mode Shape Curvature (WT-MSC) yielded the most distinct damage identification results among all evaluated metrics, effectively addressing the signal-to-noise ratio challenges inherent in optical modal analysis. As illustrated in Fig. 14, this hybrid approach creates a synergy between the high sensitivity of curvature-based metrics and the robust noise-rejection properties of wavelet analysis. A critical limitation observed in the standard Mode Shape Curvature analysis (Section 4.6) was the amplification of high-frequency experimental noise during numerical differentiation. This



(a) Δ 7.30 Hz (b) Δ 13.69 Hz (c) Δ 17.34 vs 18.25 Hz (d) Δ 24.64 vs 22.82 Hz

(e) Δ 29.20 vs 30.12 Hz (f) Δ 37.42 vs 42.89 Hz (g) Δ 46.54 vs 42.89 Hz (h) Δ 55.67 Hz

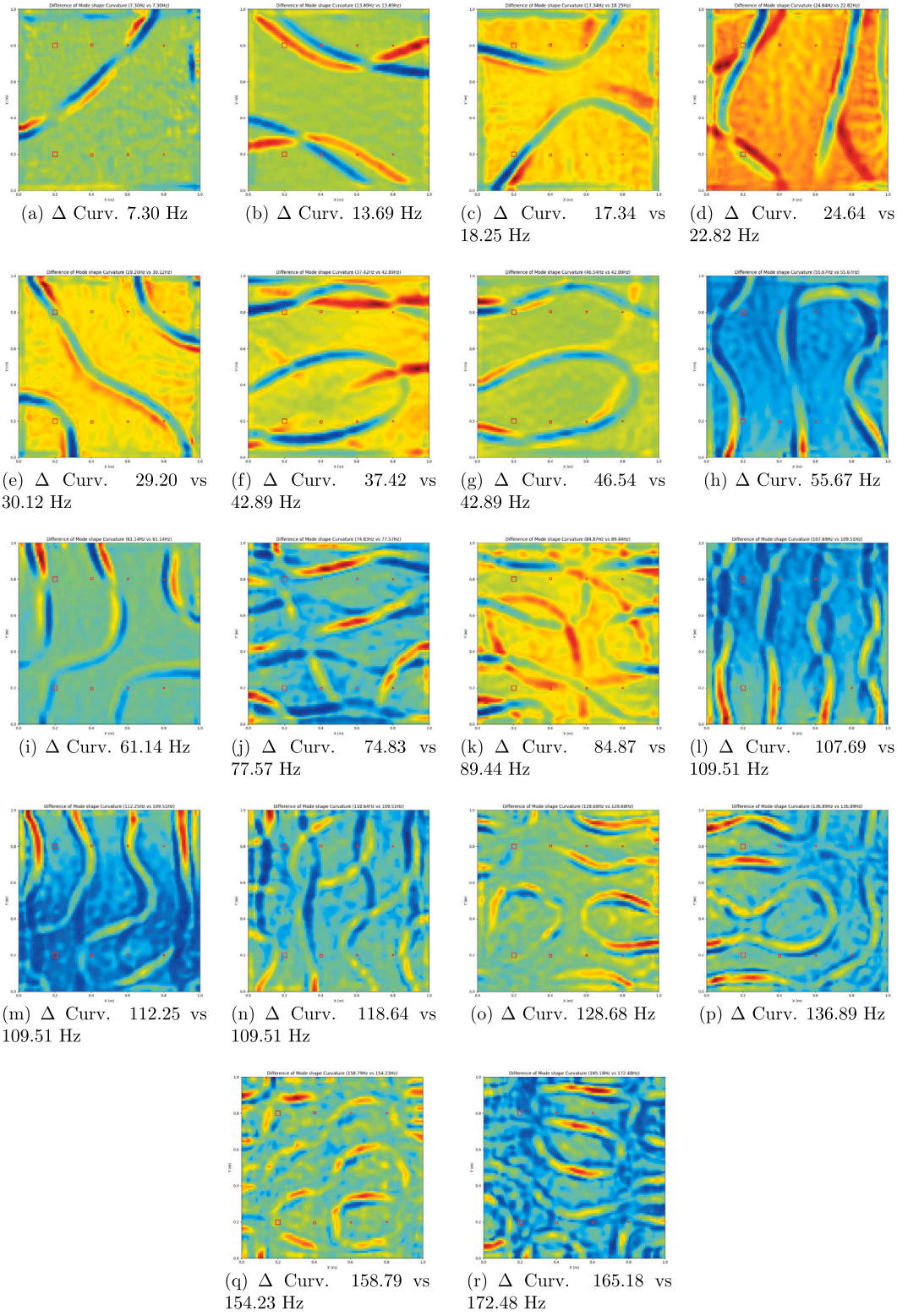
(i) Δ 61.14 Hz (j) Δ 74.83 vs 77.57 Hz (k) Δ 84.87 vs 89.44 Hz (l) Δ 107.69 vs 109.51 Hz

(m) Δ 112.25 vs 109.51 Hz (n) Δ 118.64 vs 109.51 Hz (o) Δ 128.68 Hz (p) Δ 136.89 Hz

(q) Δ 158.79 vs 154.23 Hz (r) Δ 165.18 vs 172.48 Hz

**Fig. 9.** Mode Shape Difference maps comparing the pristine (Plate 1) and damaged (Plate 2) operational deflection shapes. Captions indicate the specific frequency pair (Undamaged vs. Damaged) used for the subtraction.

noise often manifested as chaotic "static" that obscured the localized peaks caused by damage. However, by applying the Continuous Wavelet Transform (CWT) using the Mexican Hat wavelet to these curvature fields, the noise floor is significantly suppressed. The wavelet convolution functions as a spatial filter, smoothing out incoherent measurement artifacts while resonating with the coherent singularities induced by the delaminations. This improvement is visually evident when comparing the raw curvature maps to the WT-MSC scalograms. In the frequency range of 30 to 60 Hz (e.g., Fig. 14e-h), the WT-MSC method successfully isolates the upper row of defects, appearing as distinct, high-magnitude red zones against a dark blue (low energy) background. Unlike the Mode Shape Difference method, which only captured global stiffness deviations, the WT-MSC pinpoints localized discontinuities. Furthermore, unlike the raw Curvature method, it



(a) Δ Curv. 7.30 Hz (b) Δ Curv. 13.69 Hz (c) Δ Curv. 17.34 vs 18.25 Hz (d) Δ Curv. 24.64 vs 22.82 Hz

(e) Δ Curv. 29.20 vs 30.12 Hz (f) Δ Curv. 37.42 vs 42.89 Hz (g) Δ Curv. 46.54 vs 42.89 Hz (h) Δ Curv. 55.67 Hz

(i) Δ Curv. 61.14 Hz (j) Δ Curv. 74.83 vs 77.57 Hz (k) Δ Curv. 84.87 vs 89.44 Hz (l) Δ Curv. 107.69 vs 109.51 Hz

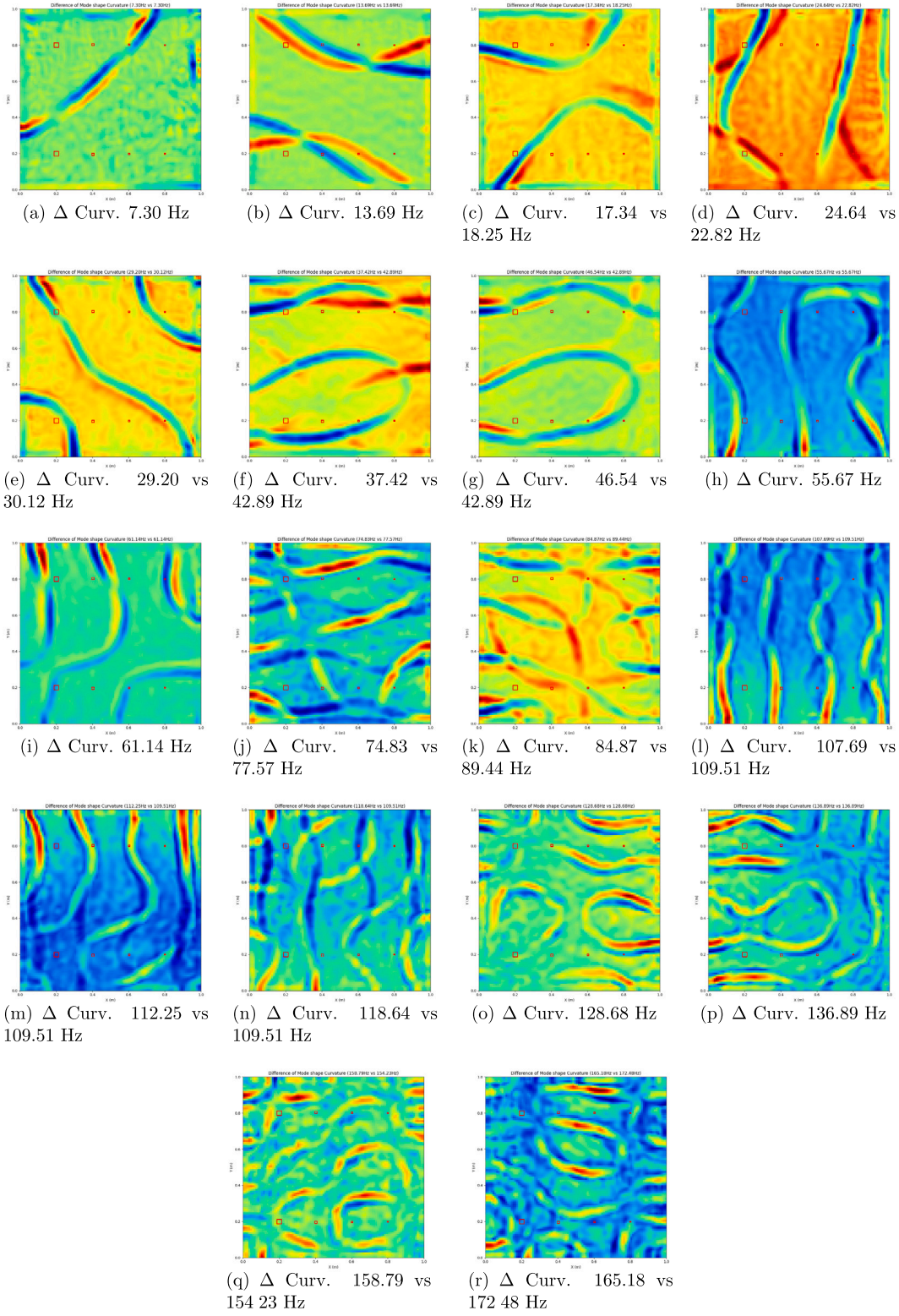
(m) Δ Curv. 112.25 vs 109.51 Hz (n) Δ Curv. 118.64 vs 109.51 Hz (o) Δ Curv. 128.68 Hz (p) Δ Curv. 136.89 Hz

(q) Δ Curv. 158.79 vs 154.23 Hz (r) Δ Curv. 165.18 vs 172.48 Hz

**Fig. 10.** Mode Shape Curvature (MSC) difference maps. These plots represent the second spatial derivative of the mode shapes, enhancing the identification of local stiffness changes.

provides a clean, interpretable image where the damage signature is distinguishable from the global modal topology. Consequently, the WT-MSC stands out as the most promising technique for reference-free inspection. It overcomes the "noise barrier" that typically renders second-derivative methods impractical for experimental data, confirming that higher-order signal processing is essential for detecting small-scale delaminations in large composite structures using full-field optical data.

However, as the frequency increases and the modal wavelength decreases, subtle irregularities begin to emerge within the damage zones. For instance, in the 109.51 Hz mode (Fig. 12l), disturbances in the smooth curvature surface coincide with the upper row of defects.



(a) Δ Curv. 7.30 Hz (b) Δ Curv. 13.69 Hz (c) Δ Curv. 17.34 vs 18.25 Hz (d) Δ Curv. 24.64 vs 22.82 Hz

(e) Δ Curv. 29.20 vs 30.12 Hz (f) Δ Curv. 37.42 vs 42.89 Hz (g) Δ Curv. 46.54 vs 42.89 Hz (h) Δ Curv. 55.67 Hz

(i) Δ Curv. 61.14 Hz (j) Δ Curv. 74.83 vs 77.57 Hz (k) Δ Curv. 84.87 vs 89.44 Hz (l) Δ Curv. 107.69 vs 109.51 Hz

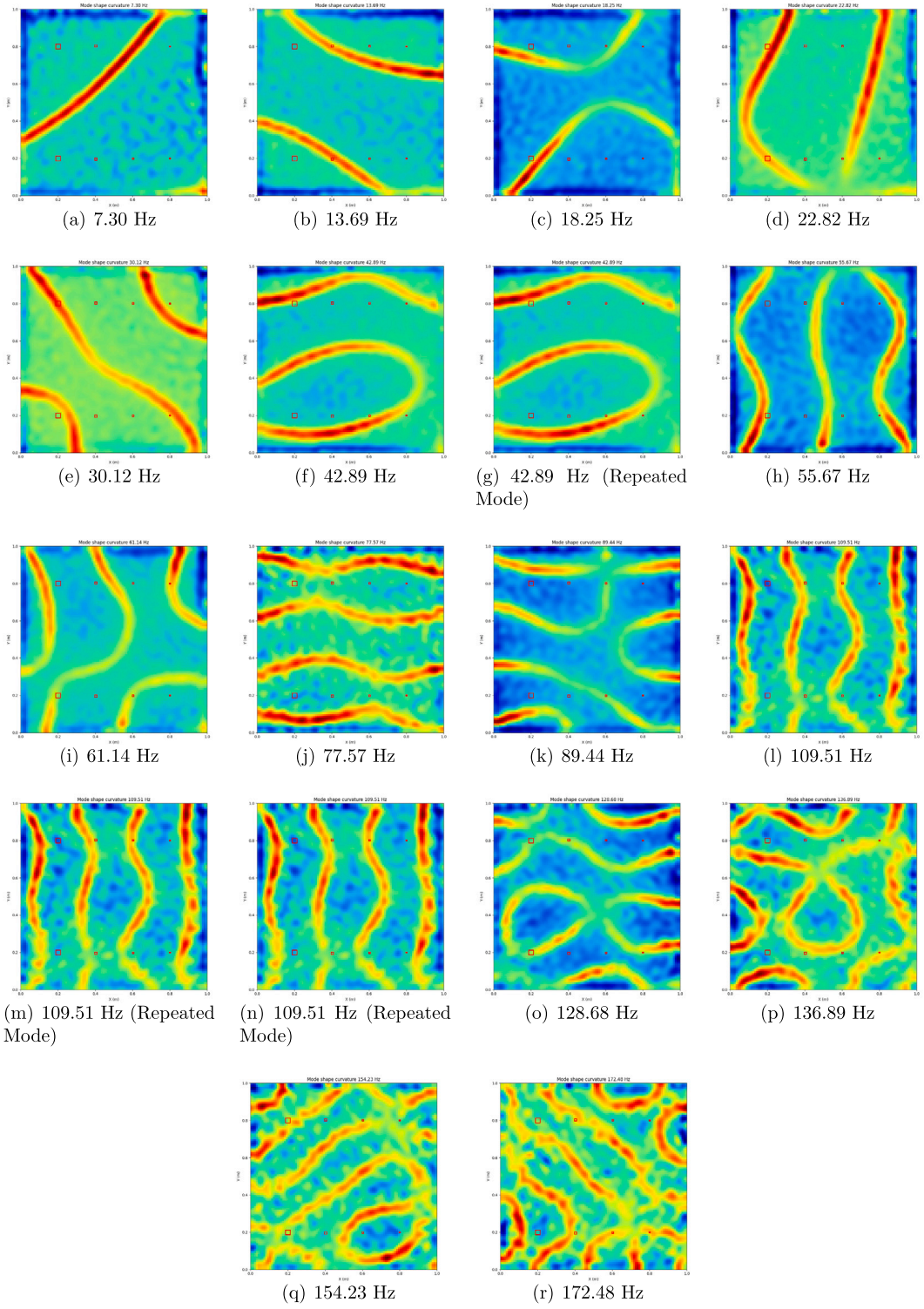
(m) Δ Curv. 112.25 vs 109.51 Hz (n) Δ Curv. 118.64 vs 109.51 Hz (o) Δ Curv. 128.68 Hz (p) Δ Curv. 136.89 Hz

(q) Δ Curv. 158.79 vs 154 23 Hz (r) Δ Curv. 165.18 vs 172 48 Hz

**Fig. 11.** Curvature of Difference maps. This metric calculates the curvature of the residual displacement field (Undamaged–Damaged), highlighting localized variations in slope change.

Nevertheless, distinguishing these damage-induced anomalies from the natural inflection points of the mode shape remains difficult. This result underscores the limitation of simple finite-difference curvature for reference-free inspection and motivates the application of the Continuous Wavelet Transform (CWT) to enhance feature extraction in the spatial domain.

To objectively evaluate the efficacy of the proposed damage identification methodology, a comprehensive quantitative analysis was conducted. This assessment moves beyond visual inspection by treating the damage detection problem as a binary classification task, where each pixel in the analyzed grid is classified as either “healthy” or “damaged” based on the computed damage index.



(a) 7.30 Hz (b) 13.69 Hz (c) 18.25 Hz (d) 22.82 Hz

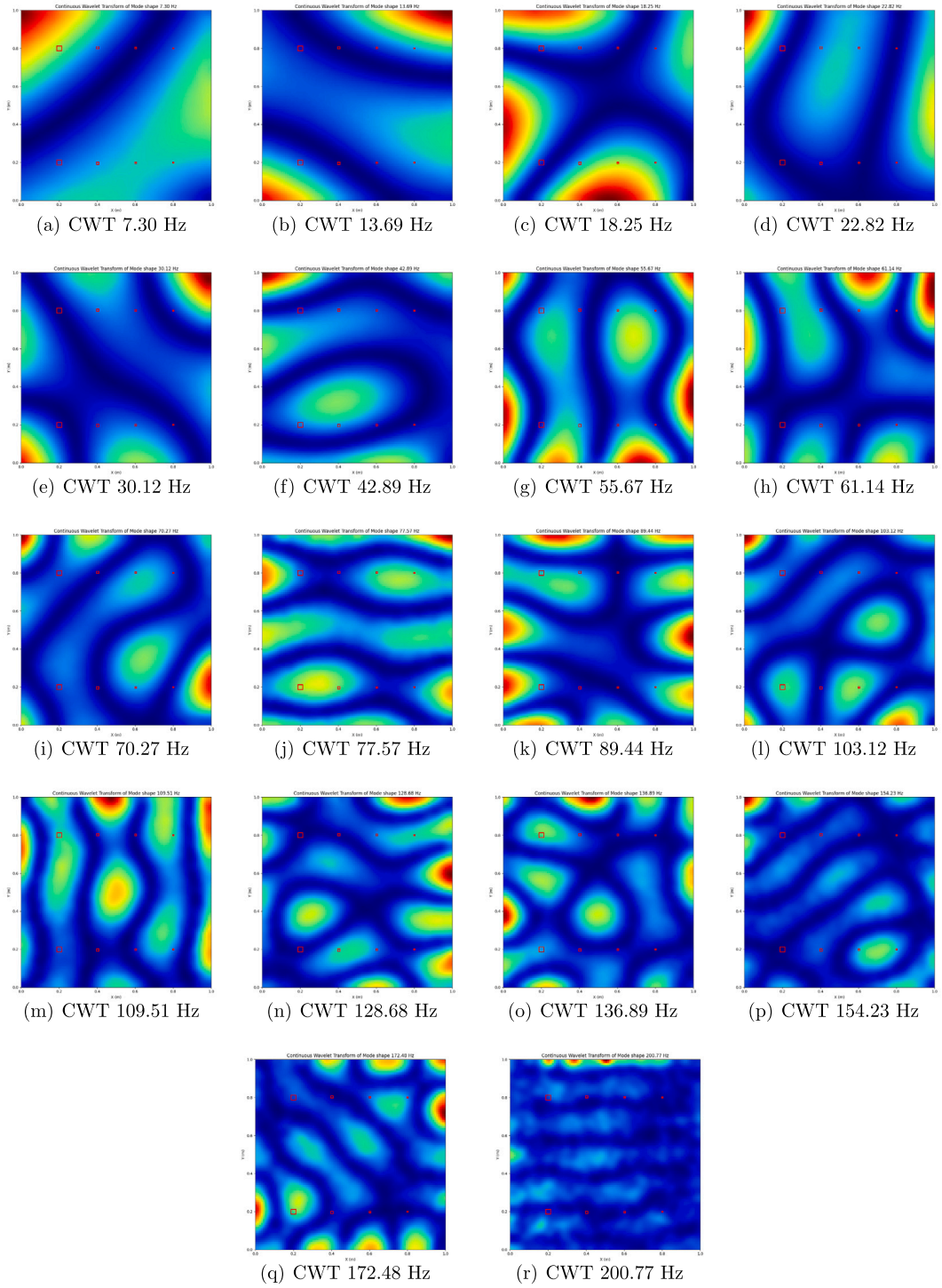
(e) 30.12 Hz (f) 42.89 Hz (g) 42.89 Hz (Repeated Mode) (h) 55.67 Hz

(i) 61.14 Hz (j) 77.57 Hz (k) 89.44 Hz (l) 109.51 Hz

(m) 109.51 Hz (Repeated Mode) (n) 109.51 Hz (Repeated Mode) (o) 128.68 Hz (p) 136.89 Hz

(q) 154.23 Hz (r) 172.48 Hz

**Fig. 12.** Mode Shape Curvature (MSC) maps for the damaged plate (Plate 2) without baseline subtraction. Red rectangles indicate the known locations of the induced damage. Anomalies in the curvature surface within these regions demonstrate the potential for reference-free damage detection.



(a) CWT 7.30 Hz (b) CWT 13.69 Hz (c) CWT 18.25 Hz (d) CWT 22.82 Hz

(e) CWT 30.12 Hz (f) CWT 42.89 Hz (g) CWT 55.67 Hz (h) CWT 61.14 Hz

(i) CWT 70.27 Hz (j) CWT 77.57 Hz (k) CWT 89.44 Hz (l) CWT 103.12 Hz

(m) CWT 109.51 Hz (n) CWT 128.68 Hz (o) CWT 136.89 Hz (p) CWT 154.23 Hz

(q) CWT 172.48 Hz (r) CWT 200.77 Hz

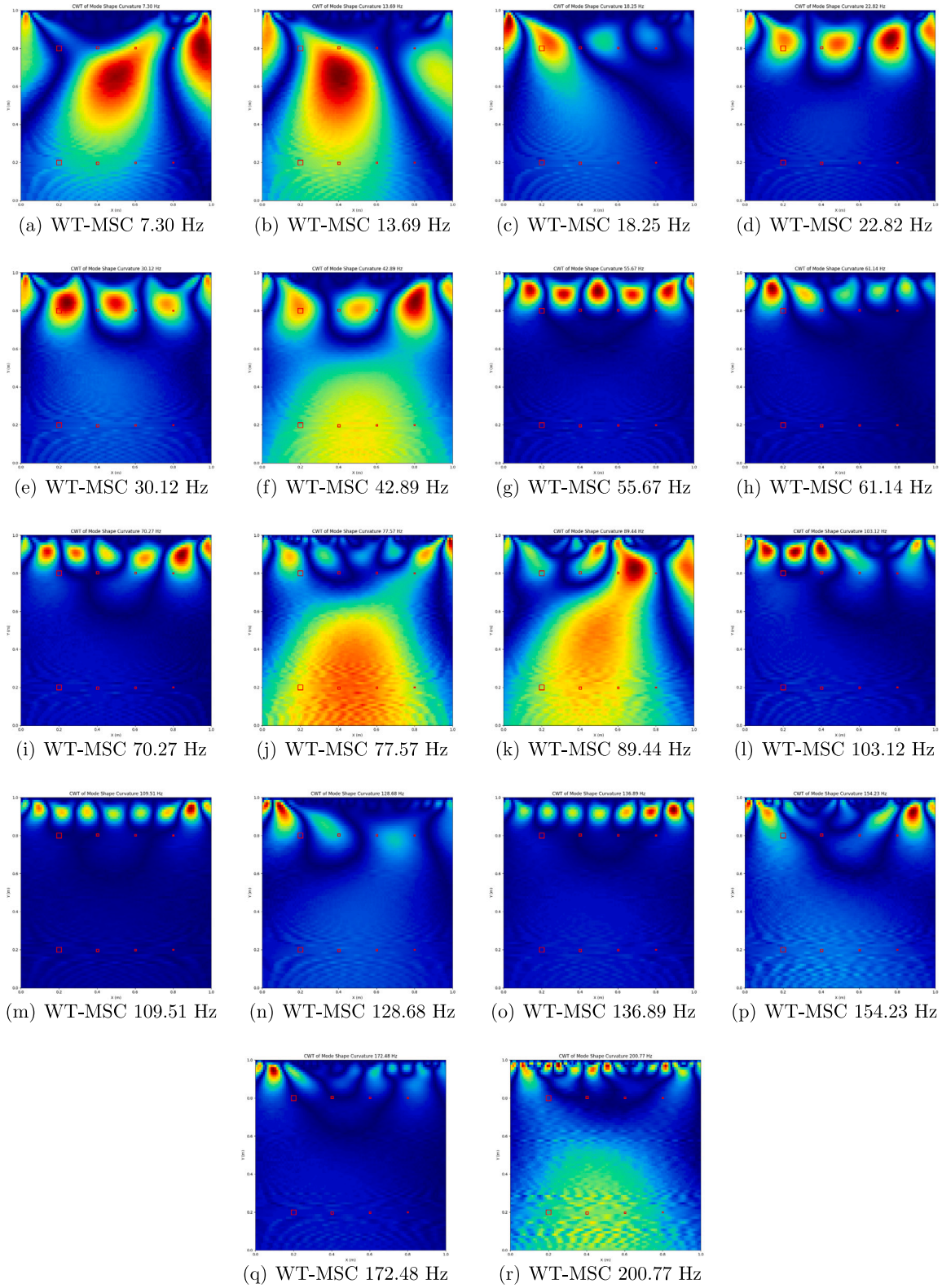
**Fig. 13.** Spatial maps of the Continuous Wavelet Transform (CWT) coefficients using the Mexican Hat wavelet. The red rectangles mark the known damage locations, which ideally correspond to high-magnitude wavelet coefficients (singularities).

To objectively assess the performance of the evaluated damage detection strategies, several quantitative statistical indicators were employed, including the Modal Assurance Criterion (MAC), Signal-to-Noise Ratio (SNR), Receiver Operating Characteristic (ROC) analysis, and the F1-Score. The mathematical formulations and definitions for these performance metrics are detailed in previous sections.

The detailed quantitative results for all 18 analyzed frequency pairs are presented in Table 2, while Table 3 provides the aggregated summary statistics (Mean ± Standard Deviation).

Analyzing the data from Table 2 reveals critical insights into the limitations of the Curvature of Difference method when applied to experimental data without advanced denoising:

- **Global Similarity (MAC):** The high MAC values for fundamental modes (e.g., 0.9872 at 7.30 Hz) indicate that the small delaminations do not significantly alter the global modal topology. However, significant drops in MAC are observed at specific resonances, such as the 37.42 Hz pair (MAC = 0.3342), suggesting



(a) WT-MSC 7.30 Hz (b) WT-MSC 13.69 Hz (c) WT-MSC 18.25 Hz (d) WT-MSC 22.82 Hz

(e) WT-MSC 30.12 Hz (f) WT-MSC 42.89 Hz (g) WT-MSC 55.67 Hz (h) WT-MSC 61.14 Hz

(i) WT-MSC 70.27 Hz (j) WT-MSC 77.57 Hz (k) WT-MSC 89.44 Hz (l) WT-MSC 103.12 Hz

(m) WT-MSC 109.51 Hz (n) WT-MSC 128.68 Hz (o) WT-MSC 136.89 Hz (p) WT-MSC 154.23 Hz

(q) WT-MSC 172.48 Hz (r) WT-MSC 200.77 Hz

**Fig. 14.** Results of the Wavelet Transform applied to the Mode Shape Curvature (WT-MSC). This method provides the highest sensitivity to damage by combining curvature analysis with wavelet-based singularity detection. Red zones indicate high wavelet coefficients correlating to damage locations.

**Table 2**
Detailed quantitative evaluation of damage detection performance for each identified mode shape pair.

| Freq. undamaged (Hz) | Freq. Damaged (Hz) | MAC (-) | SNR (dB) | ROC-AUC (-) | F1-Score (-) | Precision (-) |
|---|---|---|---|---|---|---|
| 7.30 | 7.30 | 0.9872 | −0.4815 | 0.3019 | 0.0000 | 0.0000 |
| 13.69 | 13.69 | 0.7614 | 0.3501 | 0.5604 | 0.0091 | 0.0046 |
| 17.34 | 18.25 | 0.8944 | −0.2606 | 0.4376 | 0.0025 | 0.0013 |
| 24.64 | 22.82 | 0.7888 | 0.8311 | 0.7521 | 0.0162 | 0.0083 |
| 29.20 | 30.12 | 0.9452 | 0.2335 | 0.5279 | 0.0114 | 0.0058 |
| 37.42 | 42.89 | 0.3342 | −0.1282 | 0.5153 | 0.0029 | 0.0015 |
| 46.54 | 42.89 | 0.8920 | −0.4539 | 0.4135 | 0.0000 | 0.0000 |
| 55.67 | 55.67 | 0.9007 | −0.5189 | 0.3310 | 0.0000 | 0.0000 |
| 61.14 | 61.14 | 0.8959 | −0.0795 | 0.4218 | 0.0032 | 0.0017 |
| 74.83 | 77.57 | 0.6120 | −0.7573 | 0.2211 | 0.0008 | 0.0004 |
| 84.87 | 89.44 | 0.4616 | −0.5666 | 0.3441 | 0.0000 | 0.0000 |
| 107.69 | 109.51 | 0.4402 | −0.3539 | 0.4173 | 0.0000 | 0.0000 |
| 112.25 | 109.51 | 0.8145 | −0.4310 | 0.3130 | 0.0050 | 0.0025 |
| 118.64 | 109.51 | 0.5973 | −0.1346 | 0.5195 | 0.0000 | 0.0000 |
| 128.68 | 128.68 | 0.9312 | −0.6869 | 0.2690 | 0.0000 | 0.0000 |
| 136.89 | 136.89 | 0.8623 | −0.3726 | 0.4259 | 0.0000 | 0.0000 |
| 158.79 | 154.23 | 0.8140 | −0.3721 | 0.3857 | 0.0035 | 0.0018 |
| 165.18 | 172.48 | 0.7408 | −0.6286 | 0.3092 | 0.0000 | 0.0000 |

**Table 3**
Summary of quantitative performance metrics (Mean ± standard deviation).

| Metric | Value |
|---|---|
| Mean MAC | $0.76 \pm 0.19$ |
| Mean ROC-AUC | $0.41 \pm 0.13$ |
| Mean SNR | $-0.27 \pm 0.40$ |
| Mean F1-Score | $0.003 \pm 0.005$ |
| Mean Precision | 0.002 |
| Mean Recall | 0.081 |

that certain mode shapes are more sensitive to the local stiffness variations induced by the defects.

- **Signal-to-Noise Ratio (SNR):** The predominantly negative SNR values (Mean = −0.27) quantitatively confirm that experimental noise often overwhelms the damage signal. However, exceptions exist; the 24.64 Hz vs 22.82 Hz pair exhibited a positive SNR of 0.8311, indicating that for specific resonant frequencies, the damage signature is naturally more distinct. This frequency dependence highlights the importance of analyzing a broad bandwidth.
- **Classification Performance (ROC-AUC):** The method generally struggles to automate detection, with a mean AUC of 0.41. However, the 24.64 Hz mode achieved an AUC of 0.7521, demonstrating fair classification capability. Conversely, higher frequency modes (e.g., 128.68 Hz) showed very poor performance (AUC ≈ 0.27), likely due to the amplification of noise inherent in the second-derivative calculation at higher spatial frequencies.
- **Precision and F1-Score:** The extremely low F1-scores (Mean = 0.003) reflect the difficulty of thresholding the raw curvature maps. The low precision indicates a high rate of false positives, where noise artifacts are misclassified as damage.

These quantitative results demonstrate that while simple gradient-based methods can detect anomalies in specific favorable modes (e.g., ≈ 24 Hz), they are insufficient for reliable, automated detection across the full frequency spectrum. The generally low SNR and AUC values provide a mathematical justification for the necessity of the Wavelet Transform (WT-MSC) method, which is designed to improve the SNR by leveraging multi-scale singularity detection to suppress the background noise quantified here.

To systematically summarize the capabilities and physical limitations of the proposed WT-MSC methodology, the detectability of the eight induced delaminations can be categorized into three levels and directly correlated to the defect size, depth, and the modal wavelength:

- **Detected (25.4 mm and 19.05 mm):** The largest delaminations (D1, D5 and D2, D6) were successfully and consistently identified, but primarily in higher-order modes (e.g., modes above 60 Hz). At fundamental, low-order modes, the structural wavelengths are exceptionally long, effectively "bridging" over the damage without significant local deformation. As the modal order increases, the wavelength decreases. When the wavelength approaches the spatial dimension of the defect, the interaction with the local stiffness reduction is maximized, creating a pronounced curvature anomaly. Regarding depth, the shallower defects (Top row) exhibited a slightly higher signal-to-noise ratio in the wavelet coefficients compared to the deeper ones (Bottom row), as the out-of-plane displacement gradients are more prominent closer to the surface.
- **Partially Detected (12.7 mm):** The intermediate defects (D3, D7) were only partially detected. Their signatures were entirely indistinguishable in the lower frequency range. They only began to emerge as faint, high-magnitude wavelet coefficients in the highest evaluated frequency ranges (> 100 Hz), where the spatial resolution of the mode shape curvature was sufficiently fine to capture the minimal localized strain.
- **Not Detected (6.35 mm):** The smallest defects (D4, D8) remained largely undetected across the entire evaluated frequency spectrum (up to ≈ 200 Hz). The structural wavelength, even at the highest captured modes, remained too large to be perturbed by a defect of this scale. Identifying defects of this magnitude using VBDD would require exciting the structure at kilohertz frequencies, pushing beyond the standard optical acquisition limits into the domain of ultrasonic guided waves.
- **Complex Geometries (Curved and Non-Planar Structures):** Regarding the scalability of the proposed approach to realistic aerostructures, the method remains highly feasible for curved, stiffened, or non-planar composite components. Because the data acquisition relies on a stereoscopic 3D-DIC system rather than a 2D optical setup, the cross-correlation algorithm triangulates the surface coordinates in three-dimensional space. Consequently, out-of-plane deformations are explicitly captured over complex geometric contours. Provided that a clear optical line of sight is maintained across the Region of Interest (ROI) and the lenses' depth of field is calibrated to encompass the structural curvature, the extracted 3D displacement fields can be directly subjected to the WT-MSC post-processing framework to identify localized stiffness reductions in realistic fuselage or wing components.

This relationship underscores a fundamental principle of vibration-based SHM: reliable damage detection is strictly governed by the ratio

**Table 4**
Summary of advantages and limitations for the evaluated damage detection methodologies.

| Methodology | Requirement | Key advantages | Primary limitations |
|---|---|---|---|
| Mode Shape Difference | Baseline-Dependent | - Simple computational implementation.<br>- Effectively identifies global stiffness reductions. | - Strictly requires a pristine reference structure.<br>- Low spatial resolution; struggles to pinpoint micro-defects.<br>- Sensitive to boundary condition variations between tests. |
| Raw Mode Shape Curvature (MSC) | Baseline-Free | - Eliminates the need for a pristine reference.<br>- Theoretically higher sensitivity to local stiffness drops than pure displacement. | - Catastrophically amplifies high-frequency experimental noise.<br>- Highly susceptible to false positives from interpolation artifacts.<br>- Damage signatures often obscured by the noise floor. |
| Wavelet Transform MSC (WT-MSC) | Baseline-Free | - Robust spatial filtering suppresses experimental noise.<br>- Effectively isolates and highlights damage-induced singularities.<br>- Highest sensitivity to small-scale delaminations. | - Computationally more intensive.<br>- Requires evaluation across multiple wavelet scales.<br>- Performance remains physically bounded by the modal wavelength limit (cannot detect defects drastically smaller than the wavelength). |

between the defect size and the modal wavelength. The WT-MSC method optimizes the extraction of these anomalies, but it remains fundamentally bounded by the physical deformation patterns of the excited modes.

To consolidate the comparative evaluation conducted throughout this study, Table 4 provides a concise summary of the advantages, limitations, and operational requirements associated with each evaluated damage detection methodology. This synthesis serves as a practical guideline for selecting an appropriate post-processing strategy based on the availability of baseline data and the inherent noise levels of the optical measurement system.

## 5. Conclusion

This study established a comprehensive experimental framework for the global, non-contact detection of delaminations in composite laminate structures, integrating Vibration-Based Damage Detection (VBDD) with High-Speed Three-Dimensional Digital Image Correlation (3D-DIC). The investigation systematically evaluated the efficacy of multiple damage identification metrics, ranging from algebraic mode shape differences to advanced wavelet-based curvature analysis, with the explicit goal of characterizing the proposed methodology's capabilities and limitations across three hierarchical levels of structural damage identification: damage detection (Level I), damage localization (Level II), and damage quantification (Level III).

The experimental campaign, conducted on aerospace-grade carbon fiber plates with induced delaminations, yielded the following key conclusions:

- **Viability of Optical Modal Analysis:** The 3D-DIC system successfully captured the full-field Operational Deflection Shapes (ODS) up to 200 Hz. The optical measurements demonstrated high coherence with the baseline accelerometer data, validating 3D-DIC as a robust "virtual sensor" array capable of extracting dense spatial information unavailable to single-point sensors.
- **Level I — Damage Detection:** All evaluated methods provided satisfactory evidence that damage is present in the structure. Measurable natural frequency shifts were observed across the FRF spectrum, particularly in higher-order modes, and high MAC values for fundamental modes (e.g., 0.9872 at 7.30 Hz) confirmed that small-scale delaminations reduce local stiffness without catastrophically altering the global structural topology. The accumulated evidence from frequency shifts, MAC deviations, and Mode Shape Difference maps collectively established damage existence unambiguously, demonstrating that Level I is reliably achievable with the present optical framework.
- **Level II — Damage Localization:** Significant distinctions between methods emerged at this level. Mode Shape Difference and raw Curvature Difference maps were dominated by global modal topology (antinodes) rather than the localized singularities induced by small-scale delaminations, confirming their insufficient spatial resolution for precise localization. The standard Mode Shape Curvature (MSC) method improved sensitivity to local stiffness reductions in principle, but catastrophically amplified experimental noise during numerical differentiation, as quantitatively confirmed by a mean SNR of −0.27 dB and a mean ROC-AUC of 0.41 across all evaluated mode pairs. The proposed Wavelet Transform-Mode Shape Curvature (WT-MSC) method overcame these limitations, successfully isolating damage-induced spatial singularities from background noise through multi-resolution analysis, without requiring a pristine baseline. Localization was consistently achieved for delaminations of 19.05 mm and 25.4 mm in higher-order modes (above the 6th mode, >60 Hz), where the structural wavelength is sufficiently short to interact with the localized stiffness reduction. The 12.7 mm defects were only partially detected at the highest evaluated frequencies, while the 6.35 mm delaminations remained largely unresolved within the 1–200 Hz bandwidth, establishing the current minimum detectable defect size relative to the optical acquisition limits.
- **Level III — Damage Quantification:** Full quantification of defect severity (including precise size, depth, and degradation state estimation) was not achieved within the present framework and constitutes a recognized limitation. Nevertheless, partial qualitative evidence toward quantification was observed: larger delaminations consistently produced more prominent wavelet coefficient peaks across a broader frequency range, and shallower defects (top row) exhibited marginally higher wavelet coefficients than deeper ones at equivalent sizes, consistent with the expected attenuation of out-of-plane displacement gradients with increasing interlaminar depth. Establishing a formal, calibrated relationship between wavelet coefficient intensity and defect area or depth remains an open challenge, as it constitutes an ill-posed inverse problem requiring additional constraints beyond ODS data alone.
- **Influence of Modal Order:** The detectability of defects was found to be strongly frequency-dependent across all three levels. Higher-order modes, characterized by shorter structural wavelengths and

higher local curvature gradients, exhibited significantly greater sensitivity to delaminations compared to fundamental bending modes. This observation underscores a fundamental physical principle governing VBDD: reliable damage detection and localization are strictly bounded by the ratio between the defect size and the modal wavelength, irrespective of the post-processing strategy employed.

In summary, this work validates the integration of 3D-DIC with WT-MSC as a promising non-destructive testing strategy, with demonstrated capability at Levels I and II of the damage identification hierarchy. Level III quantification remains an open challenge within the optical modal analysis paradigm. Future research will focus on: establishing calibration curves between wavelet coefficient intensity and defect area for controlled specimen geometries; integrating model-updating frameworks to enable quantitative severity estimation; extending the frequency bandwidth toward the kilohertz range to improve sensitivity to sub-centimeter defects; and exploring machine learning integration to automate damage signature classification in complex aerostructural geometries.

## CRediT authorship contribution statement

**Matheus Teixeira de Moura Santos:** Writing – original draft. **Guilherme Ferreira Gomes:** Writing – review & editing, Supervision, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgments

The authors would like to acknowledge the financial support from the Brazilian agencies: CAPES (Coordenação de Aperfeiçoamento de Pessoal de Nível Superior), CNPq (Conselho Nacional de Desenvolvimento Científico e Tecnológico - 405598/2022-0) and FAPEMIG (Fundação de Amparo à Pesquisa do Estado de Minas Gerais - APQ-02290-23 and APQ-00062-24).

## Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.compstruct.2026.120506.

## Data availability

Data will be made available on request.

## References

[1] Gomes GF, de Almeida FA, da Cunha Jr. SS, Ancelotti Jr. AC. An estimate of the location of multiple delaminations on aeronautical CFRP plates using modal data inverse problem. Int J Adv Manuf Technol 2018;99(5):1155–74.

[2] Greenhalgh ES. Failure Analysis and Fractography of Polymer Composites. Woodhead Publishing; 2009, ch. 4.

[3] Federal Aviation Administration. Visual inspection for aircraft. Advis Circ 1997;AC:43–204.

[4] Wang S, Abbott TB, Fong R-Y, Xu C, Yuan F-G. Damage detection using digital image correlation and stereo cameras. Struct Health Monit 2024;23(1):294–308.

[5] Gomes GF, Chaves JAS, de Almeida FA. An inverse damage location problem applied to AS-350 rotor blades using bat optimization algorithm and multiaxial vibration data. Mech Syst Signal Process 2020;145:106932.

[6] de Souza Mello FM, Pereira JLJ, Gomes GF. Multi-objective sensor placement optimization in SHM systems with Kriging-based mode shape interpolation. J Sound Vib 2024;568:118050.

[7] Doebling SW, Farrar CR, Prime MB. A summary review of vibration-based damage identification methods. Shock Vib Dig 1998;30:91–105.

[8] Janeliukstis R, Chen X. Review of digital image correlation application to large-scale composite structure testing. Compos Struct 2021;271:114143.

[9] Sutton MA, Orteu J-J, Schreier H. Image correlation for shape, motion and deformation measurements: basic concepts, theory and applications. New York, NY: Springer Science & Business Media; 2009.

[10] Khalid S, Kim H-S, Kim HS, Choi J-H. Inspection interval optimization for aircraft composite tail wing structure using numerical-analysis-based approach. Mathematics 2022;10(20):3836.

[11] Wang W, Xue Y, He C, Zhao Y. Review of the typical damage and damage-detection methods of large wind turbine blades. Energies 2022;15(15):5672.

[12] Riccio A. Delamination in the context of composite structural design. In: Delamination behaviour of composites. Elsevier; 2008, p. 28–64.

[13] Zou Y, Tong L, Steven GP. Vibration-based model-dependent damage (delamination) identification and health monitoring for composite structures — a review. J Sound Vib 2000;230(2):357–78.

[14] Ratcliffe CP. Damage detection using a modified Laplacian operator on mode shape data. J Sound Vib 1997;204(3):505–17.

[15] Wu D, Law SS. Damage localization in plate structures from uniform load surface curvature. J Sound Vib 2004;276(1–2):227–44.

[16] Pandey A, Biswas M, Samman M. Damage detection from changes in curvature mode shapes. J Sound Vib 1991;145(2):321–32.

[17] Xu W, Cao M, Ostachowicz W, et al. Two-dimensional curvature mode shape method based on wavelets and teager energy for damage detection in plates. J Sound Vib 2015;347:266–78.

[18] Abdo MA-B. Damage detection in plate-like structures using high-order mode shape derivatives. Int J Civ Struct Eng 2012;2:801–16.

[19] Dos Santos JVA, Lopes HMR, Vaz M, et al. Damage localization in laminated composite plates using mode shapes measured by pulsed TV holography. Compos Struct 2006;76(3):272–81.

[20] Cao M, Xu W, Ostachowicz W, et al. Damage identification for beams in noisy conditions based on Teager energy operator-wavelet transform modal curvature. J Sound Vib 2014;333(6):1543–53.

[21] Schmidt T, Tyson J, Galanulis K. Full-field dynamic displacement and strain measurement using advanced 3D image correlation photogrammetry: Part 1. Exp Tech 2003;27(3):47–50.

[22] Helfrick MN, Niezrecki C, Avitabile P, et al. 3D digital image correlation methods for full-field vibration measurement. Mech Syst Signal Process 2011;25(3):917–27.

[23] Helfrick M, Niezrecki C, Avitabile P. Curvature methods of damage detection using digital image correlation. In: Proceedings of SPIE — sensors and smart structures technologies for civil, mechanical, and aerospace systems, vol. 7295, SPIE; 2009, p. 72950D.

[24] Clough RW. Finite element stiffness matricess for analysis of plate bending. In: Proc. of the first conf. on matrix methods in struct. mech. 1965, p. 515–46.